Applied Sciences and Humanities

Unit 1 Matrices

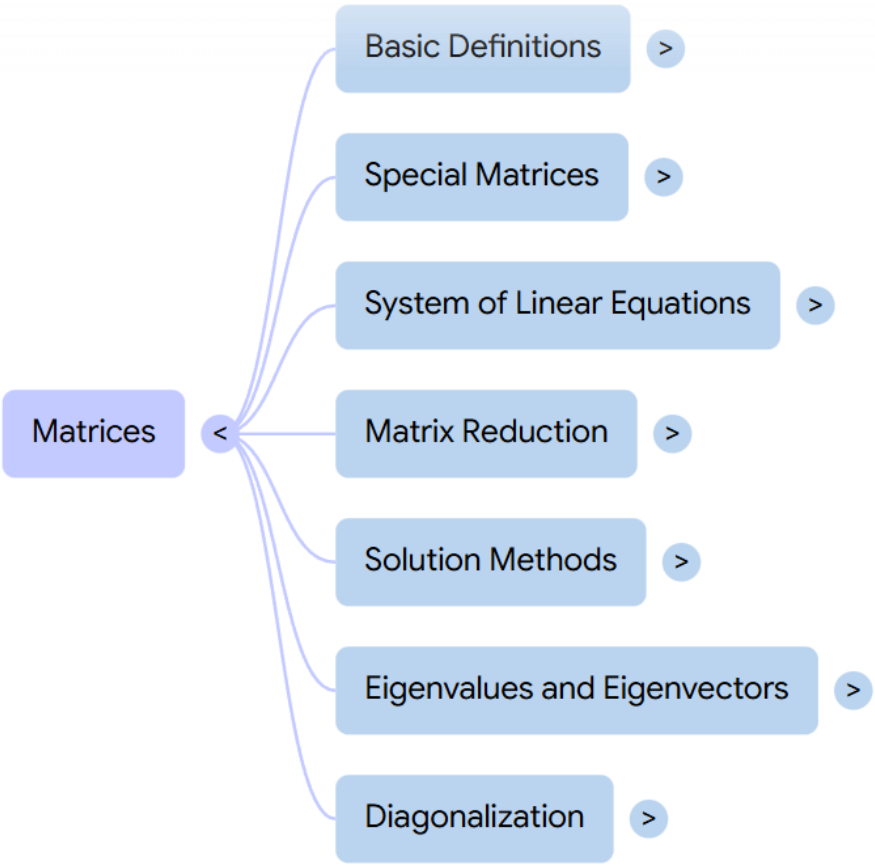
Study Guide

Dr. Monika Naik || Ms. Monali Darji
ASH, PIET
Parul University

INDEX

Sr. No.	Topics	Subtopics	Page No.
1.0	**Matrices**	Definition	4
1.1	**Basis of matrices and its types**	Trace, Symmetric, Skew-symmetric, singular and non-singular matrix, Orthogonal matrix	4-6
1.2	**System of Linear Equations**	Definition	7
1.2.1	Linear Equations	Definition	7
1.2.2	System of Linear Equations	Types of system of Linear equations, Geometrical Interpretation of system of Linear Equations	7-9
1.2.3	Augmented Matrix	Definition	7
1.2.4	Condition of Consistency for non-homogeneous system	Types of Solutions	7
1.3	**Row Echelon Form and Reduced Row Echelon Form of a matrix**	Definition	7
1.4	**Methods of Solving system of Linear Equations**	-	7
1.4.1	Gauss Elimination Method	Method Explanation and Examples	10
1.4.2	LU Decomposition Method	Method Explanation and Examples	22
1.5	**Homogeneous Equations**	Method Explanation and Examples	17
1.6	**Rank of a Matrix**	Definition	22
1.6.1	Methods for finding Rank of a matrix	By Determinant Method By Row echelon Form	22-29
1.7	**Eigenvalues and Eigen vectors**	Definition and Results	29-32

1.8	**Algebraic Multiplicity and Geometric Multiplicity**	Definition and Examples	33-36
1.9	**Cayley Hamiltonian Theorem**	Statement and Examples	37-39
1.10	**Diagonalization**	Definition and Examples	39-41
1.11	**Application of Matrices**	-	41
1.11.1	Coding and decoding	Examples	41-44
1.11.2	Principal Component Analysis	Definition and Examples	44-47
—	**References**	Standard textbooks and datasheets cited	48



Road map of matrices

1.0 Matrix:

A Matrix is a rectangular array of numbers (or functions) enclosed in brackets. These number or functions are called entries or elements of the matrix.

e.g.

$$\begin{bmatrix} 4 & -2 & 1 \\ 0 & 3 & 5 \end{bmatrix}, \begin{bmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{bmatrix}$$

Operations on Matrices:

Addition of two matrices:

One of the basic operations that can be performed on matrices is the addition operation. Just as we add two or more integers, two or more matrices can also be added in a similar way.

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{bmatrix}$$

Multiplication by a scalar

If $A = [a_{ij}]_{m \times n}$ is a matrix and k is a scalar, then kA is another matrix which is obtained by multiplying each element of A by the scalar k. In other words, $kA = k\,[a_{ij}]_{m\times n} = [k\,(a_{ij})]_{m\times n}$, that is, (i, j)th element of kA is ka_{ij} for all possible values of i and j.

Multiplication of two matrices:

The product of two matrices A and B is defined if the number of columns of A is equal to the number of rows of B. Let A = $[a_{ij}]$ be an m × n matrix and B = $[b_{jk}]$ be an n × p matrix. Then the product of the matrices A and B is the matrix C of order m × p. To get the $(i, k)^{th}$ element c of the matrix C, we take the ith row of A and k^{th} column of B, multiply them element-wise and take the sum of all these products.

In other words, if A = $[a_{ij}]m{\times}n$, B = $[b_{jk}]n{\times}p$, then the ith row of A = $[a_{i1}\ a_{i2} \ldots a_{in}]$ and the k_{th} column of then, $c = a_{i1}b1k + a_{i2}b_{2k} + a_{i3}b_{3k} + \ldots + a_{in}b_{nk} = \sum a_{ij}b_{jk}$. The matrix C = $\Sigma[c_{ik}]_{m\times p}$ is the product of A and B.

1.1 Basic of Matrix and its types

Row Matrix:

A matrix having only one row and any number of columns, is called a row matrix or row vector.

e.g.: $A = \begin{bmatrix} 2 & 5 & -3 & 4 \end{bmatrix}$

Column Matrix:

A matrix, having only one column and any number of rows, is called a column matrix or column vector.

e.g.: $A = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$

Zero or Null Matrix:

A matrix, whose all the elements are zero, is called a zero or null matrix and is denoted by **0**.

e.g. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Square Matrix:

A matrix, in which the number of rows is equal to the number of columns, is called a square matrix,

e.g. $\begin{bmatrix} 1 & 3 & 2 \\ -1 & 4 & -5 \\ 2 & 6 & 8 \end{bmatrix}$

Diagonal Matix:

A square matrix, all of whose non-diagonal elements are zero and at least one diagonal element is non-zero, is called a diagonal matrix.

e.g. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}$

Unit or Identity Matix:

A diagonal matrix, all whose diagonal elements are unity, is called a unit or identity matrix and is denoted by I,

e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Scalar Matrix:

A diagonal matrix, all of whose elements are equal, is called a scalar matrix,

e.g. $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Upper Triangular Matrix:

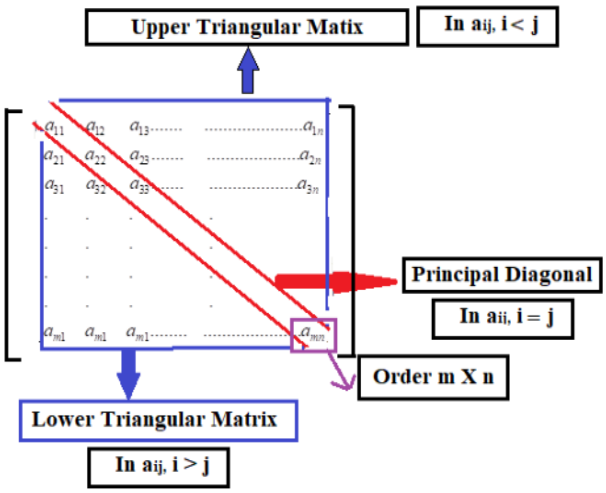
A square matrix, in which all the elements below the diagonal are zero, is called an upper triangular matrix.

e.g. $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 4 & -5 \\ 0 & 0 & 8 \end{bmatrix}$

Lower Triangular Matrix:

A square matrix, in which all the elements above the diagonal are zero, is called a lower triangular matrix.

e.g. $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 4 & 6 & 8 \end{bmatrix}$



Trace of a matrix:

If A is a square matrix, the trace of A, denoted by $tr(A)$ and is defined to be the sum of entries on the main diagonal of A. The trace of A is undefined if A is not a square matrix.

e.g. If $A = \begin{bmatrix} 4 & 5 \\ 10 & 6 \end{bmatrix}$, then $tr(A) = 4 + 6 = 10$.

Transpose of a Matrix:

A matrix obtained by interchanging rows and columns of a matrix is called transpose of a matrix and is denoted by A^T *or* A'.

If $A = [a_{ij}]_{m \times n}$, *then* $A^T = [a_{ji}]_{n \times m}$ e.g. $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ -4 & 1 \end{bmatrix}$ *then*, $A^T = \begin{bmatrix} 1 & 0 & -4 \\ -1 & 2 & 1 \end{bmatrix}$

Determinant of a matrix

If A is a square matrix, then determinant of A is represented as $|A|$ or $det(A)$.

$$|A| = \begin{vmatrix} 1 & 3 & 2 \\ -1 & 4 & -5 \\ 2 & 6 & 8 \end{vmatrix}$$

Symmetric matrix: - For any square matrix A, if $A = A^T$ then, it is known as symmetric matrix.

Skew-symmetric matrix: - For any square matrix A, if $A = -A^T$ then it is known as Skew symmetric matrix.

Singular and non-singular matrix: -

For any square matrix A, if $|A| \neq 0$, then it is known as non-singular matrix and if $|A| = 0$ then it is known as singular matrix.

e.g.

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 4 \end{bmatrix} \Rightarrow |A| = 8 - 8 = 0 \Rightarrow$$

Singular Matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow |A| = 4 - 6 = -2 \neq 0$$

$$\Rightarrow \text{Non} - \text{Singular Matrix}$$

Orthogonal Matrix:- The matrix is said to be an orthogonal matrix if the product of a matrix and its transpose give an identity value. i.e. $AA^T = I$

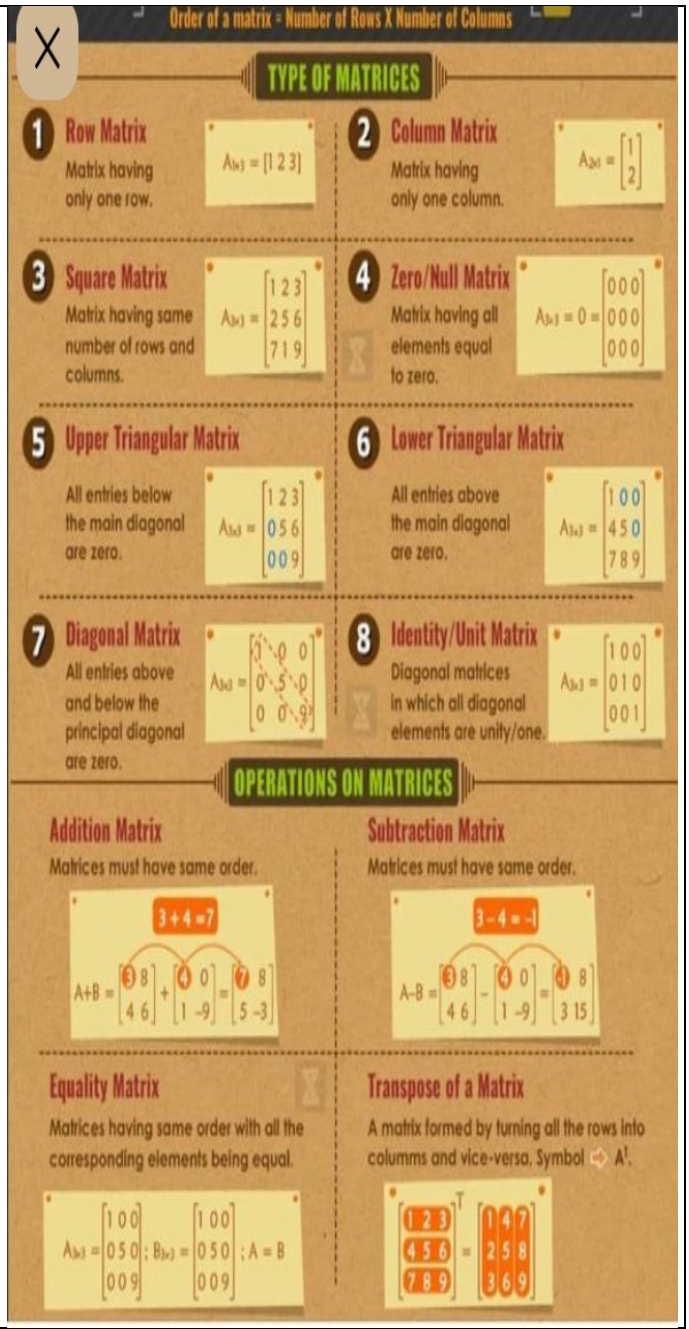
e.g.

Given A is an orthogonal matrix because

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$



Exercise:

1. Identify which of the following matrices are square, diagonal, upper or lower triangular, symmetric, or skew-symmetric. Find the trace of the matrices, if possible:

$$(i)\, A = \begin{bmatrix} 2 & -3 & 4 \\ -4 & 3 & -2 \\ 3 & 2 & 4 \end{bmatrix} \quad (ii)\, B = \begin{bmatrix} 1 & 3 & 6 \\ -1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \quad (iii)\, C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Are the given matrices orthogonal or not?

$$A=\begin{bmatrix}2 & 2 & 1\\ -2 & 1 & 2\\ 1 & -2 & 2\end{bmatrix},\ B=\begin{bmatrix}\text{Cos}\,\theta & 0 & \text{Sin}\,\theta\\ 0 & 1 & 0\\ -\text{Sin}\,\theta & 0 & \text{Cos}\,\theta\end{bmatrix},\ C=\begin{bmatrix}1 & -2\\ 2 & 1\end{bmatrix},\ D=\frac{1}{3}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ -2 & 2 & 1\end{bmatrix},E=\begin{bmatrix}1 & 3 & 1\\ 3 & 1 & 2\\ 1 & 2 & 2\end{bmatrix}$$

3. Find a,b,c if $A=\begin{bmatrix}\frac{1}{3} & \frac{2}{3} & a\\ \frac{2}{3} & \frac{1}{3} & b\\ \frac{2}{3} & -\frac{2}{3} & c\end{bmatrix}$ is orthogonal.

<u>Elementary Transformations:</u>

Elementary transformation is any one of the following operations on a matrix.

(i) The interchange of any two rows (or columns)
R_{ij} : Interchange of ith and jth row.

(ii) The multiplication of the elements of any row (or column) by any non-zero number.
kR_i : Multiplication of ith row by a non-zero number k.

(iii) Addition of k times the jth row to the ith row
$R_i + kR_j$: Addition of k times the jth row to the ith row.

<u>1.2 System of Linear Equations:</u>

<u>1.2.1 Linear Equations</u>: Any straight line in the xy-plane can be represented algebraically by equation of the form $ax + by = c$, where $a\&b$ are real numbers.

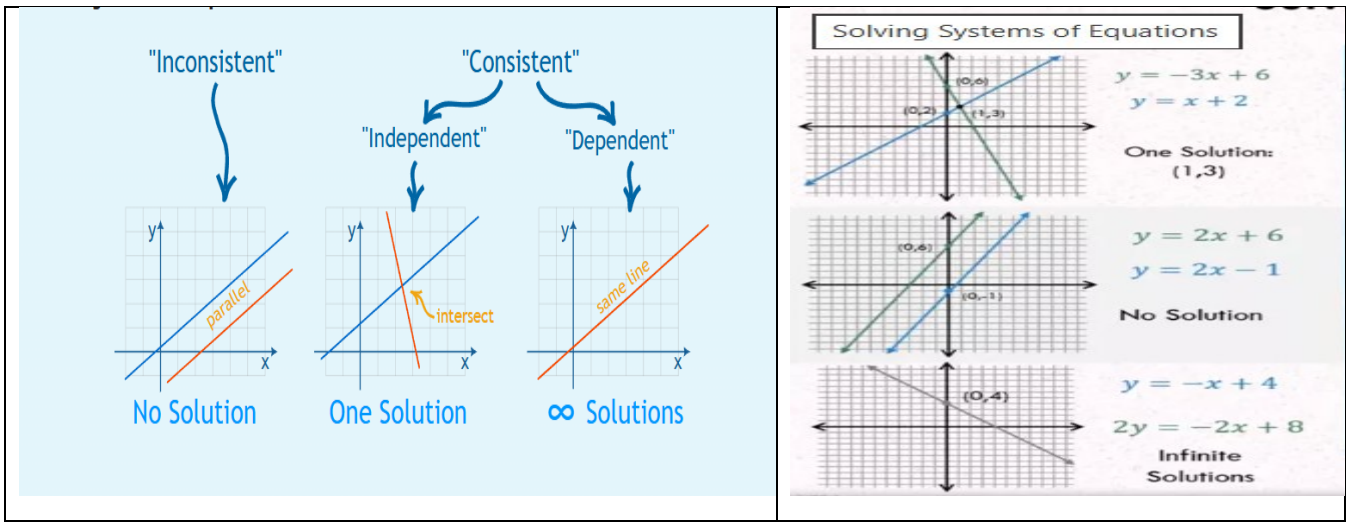
<u>1.2.2 System of Linear Equations:</u>

A **<u>system of linear equation</u>** is a collection of one or more linear equations involving the same variables. A linear system of m linear equations in n variables: An arbitrary system of m linear equations in n variables $x_1, x_2, x_3, \ldots\ldots x_n$ is a set of equations of the form

$$\sum_{j=1}^{n} a_{ij}x_j = b_i\ (i = 1,2,3,\ldots,m, j = 1,2,3,\ldots,n)$$

<u>A system of linear equations has either</u>

1. No solutions, or
2. Exactly one solution, or
3. Infinitely many solutions

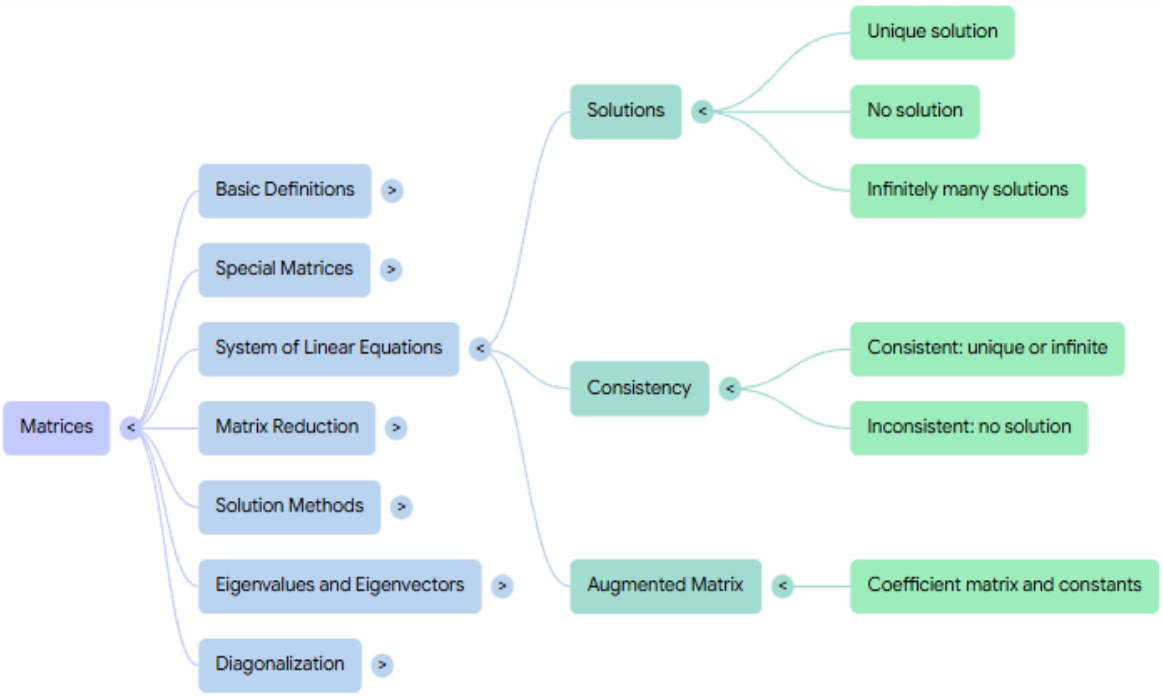


Geometrical interpretation of system of Linear equations

Remarks:

(i) The system is said to be consistent if we get infinitely many solutions or unique solution.

(ii) The system is said to be inconsistent if we get No solution.



Schematic diagram for system of Linear equations

1.2.3 Augmented matrix:

A system of m equations in n unknowns can be abbreviated by writing only the rectangular array of numbers.

$$\begin{bmatrix} a_{11}a_{12}\cdots a_{1n}|b_1 \\ a_{21}a_{22}\cdots a_{2n}|b_2 \\ \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\ a_{m1}a_{m2}\ldots a_{mn}|b_m \end{bmatrix}$$

This is known as augmented matrix.

e.g.

Find the augmented matrix for each of the following system of linear equations:

$$2x_1 + \quad +2x_3 = 1$$
$$3x_1 - x_2 + 4x_3 = 7$$
$$6x_1 + x_2 - x_3 = 0$$

Then, augmented matrix is given by $\begin{bmatrix} 2 & 0 & 2 & |1 \\ 3 & -1 & 4 & |7 \\ 6 & 1 & -1 & |0 \end{bmatrix}$.

1.2.4 Condition of Consistency for non-homogeneous system:

(1) If there is a zero row to left of the augmentation bar but the last entry of this row is non-zero then the system has **no solution**.

e.g.: $\begin{bmatrix} 1 & 0 & 2|1 \\ 0 & 1 & 4|7 \\ \mathbf{0} & \mathbf{0} & \mathbf{0|4} \end{bmatrix}$

(2) If at least one of the columns on the left of the augmentation bar has zero element pivot entry, then the system has **infinitely many solutions**.

e.g.: $\begin{bmatrix} 1 & 0 & 2|1 \\ 0 & 1 & 4|7 \\ \mathbf{0} & \mathbf{0} & \mathbf{0|0} \end{bmatrix}$

(3) If all the rows having the leading entry 1 then the system has **unique solution.**

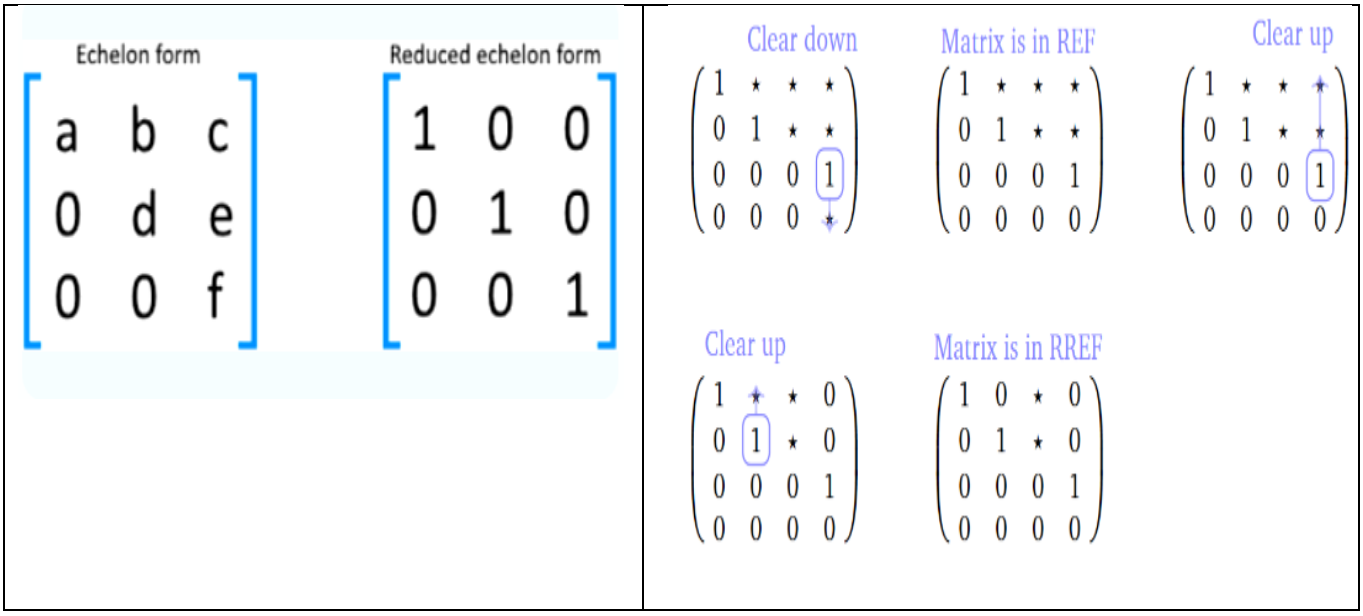
e.g.: $\begin{bmatrix} 1 & 0 & 2|1 \\ 0 & 1 & 4|7 \\ \mathbf{0} & \mathbf{0} & \mathbf{1|4} \end{bmatrix}$

1.3 Row-Echelon (RE)form and Row-Reduced Echelon (RRE) form of a matrix

Definition: A rectangular matrix is in row-echelon form (or echelon form) if it has the following three properties:

1. The **first** element in each row must be **non-zero** and equals to **1**, that is called **leading entry 1**.
2. All the **leading 1's** must be on the **right-hand** side of the matrix.
3. If any **zero row** is available, then it must be **below** to the **all-leading 1**.

If the matrix satisfies the **4th property** (i.e., In each column except leading 1 if all entries are zero) then **row-echelon form (RE form)** becomes **reduced row echelon form (RRE form)**.



Illustrations of REF and RREF

Exercise

1. Which of the following matrices are in row-echelon or echelon form?

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$ (e) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

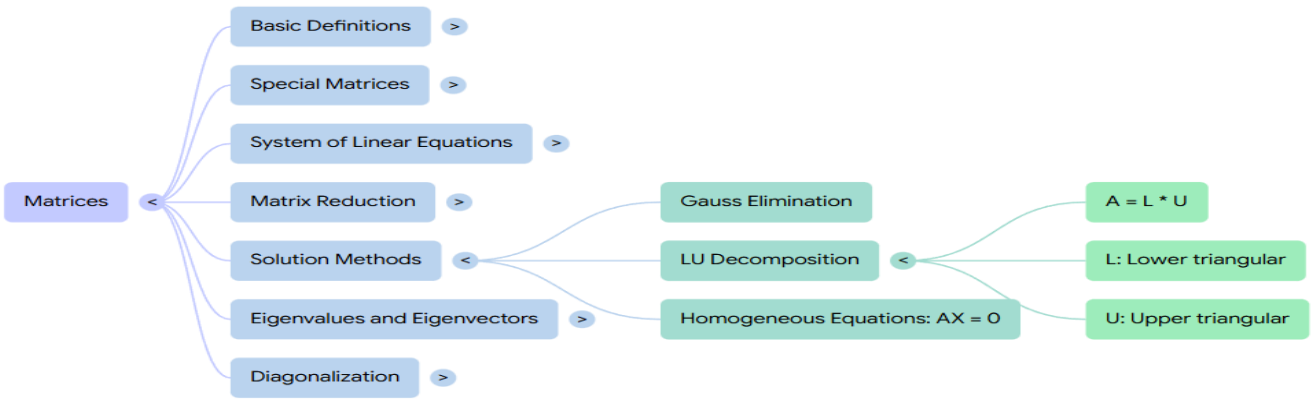
2. Which of the following matrices are in reduced row-echelon or reduced echelon form?

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (e) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (f) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (g) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

<u>1.4 Methods of solving system of linear equations:</u>

1.4.1 Gauss – Elimination Method: (Using Row Echelon Form)

1.4.2 LU Decomposition Method



Schematic diagram for Methods of Solution of system of Linear Equations

Case-1: Unique solution

Example: Solve the following system by gauss- Elimination method

$$x + y + z = 6$$
$$x + 2y + 3z = 14$$
$$2x + 4y + 7z = 30$$

Solution:

The matrix form of the given system is

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ 30 \end{bmatrix}$$

The augmented matrix is

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 4 & 7 & 30 \end{array}\right]$$

Now, to convert the given augmented matrix in row-echelon form we apply elementary operations as following.

$$R_2 \to R_2 - R_1, R_3 \to R_3 - 2R_1$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 2 & 5 & 18 \end{array}\right]$$
$$R_3 \to R_3 - 2R_2$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

The corresponding system of equation is

$$x + y + z = 6$$
$$y + 2z = 8$$
$$z = 2$$

By using back substitution of $z = 2$ in $y + 2z = 8$, we get $y = 4$ and $z = 2$ & $y = 4$ in $x + y + z = 6$ we get $x = 0$.

$x = 0, y = 4, z = 2$ is **unique solution** of given system.

Case-2: No solution

Example: Solve the following system of equation by Gauss elimination.

$$-2b + 3c = 1$$
$$3a + 6b - 3c = -2$$
$$6a + 6b + 3c = 5$$

Solution:

The matrix form of the given system is

$$\begin{bmatrix} 0 & -2 & 3 \\ 3 & 6 & -3 \\ 6 & 6 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right]$$

$$R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right]$$

$$R_1 \to (1/3)R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2/3 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right]$$

$$R_3 \to R_3 - 6R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2/3 \\ 0 & -2 & 3 & 1 \\ 0 & -6 & 9 & 9 \end{array}\right]$$

$$R_3 \to R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2/3 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 6 \end{array}\right]$$

The system of linear equation is

$a + 2b - c = -2/3$

$-2b + 3c = 1$

$0a + 0b + 0c = 6$ is not possible.

This shows that the system has **no solution**

Case-3: Infinitely many solutions

Example: Solve the following system by Gauss elimination method.

$$4x - 2y + 6z = 8$$
$$x + y - 3z = -1$$
$$15x - 3y + 9z = 21$$

Solution:

The matrix form of the given system is

$$\begin{bmatrix} 4 & -2 & 6 \\ 1 & 1 & -3 \\ 15 & -3 & 9 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -1 \\ 21 \end{bmatrix}$$

The augmented matrix is

$$[A|B] = \left[\begin{array}{ccc|c} 4 & -2 & 6 & 8 \\ 1 & 1 & -3 & -1 \\ 15 & -3 & 9 & 21 \end{array}\right]$$

$$R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 4 & -2 & 6 & 8 \\ 15 & -3 & 9 & 21 \end{array}\right]$$

$$R_2 \to R_2 - 4R_1, R_3 \to R_3 - 15R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & -18 & 54 & 36 \end{array}\right]$$

$$R_2 \to (-1/6)R_1, R_3 \to (-1/6)R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 1 & -3 & -2 \end{array}\right]$$

$$R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The corresponding system of equations is

$$x + y - 3z = -1$$
$$y - 3z = -2$$

Assigning the free variable z an arbitrary value t,
$y = 3t - 2$,

$$x = -1 - 3t + 2 + 3t = 1$$

Hence, $x = 1, y = 3t - 2, z = t$ is solution of the given system of equations.

Since t is arbitrary real number, The system has **infinitely many solutions.**

Example: Consider the following system

$$x + y + z = 6$$
$$x + 2y + 3z = 10$$
$$x + 2y + \lambda z = \mu$$

For what values of λ and μ the system has (i) infinitely many solutions (ii) unique solution and (iii) no solution.

Solution: The Augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$$

$$R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right]$$

$$R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

(i) If $\lambda - 3 = 0$ and $\mu - 10 = 0$, that is if $\lambda = 3$ and $\mu = 10$ then the system has infinitely many solutions.
(ii)If$\lambda - 3 = 0$ then the system has a unique solution. That is $\lambda \neq 3$ and μ can possess any real value.
(iii)If $\lambda - 3 = 0$ and$\mu - 10 \neq 0$, that is if $\lambda = 3$ and $\mu \neq 10$ then the system does not have any solution.

Example:
Solve the following system by gauss elimination method.

$$\frac{-1}{x} + \frac{3}{y} + \frac{4}{z} = 30$$
$$\frac{3}{x} + \frac{2}{y} - \frac{1}{z} = 9$$
$$\frac{2}{x} - \frac{1}{y} + \frac{2}{z} = 10$$

<u>Solution:</u>
Let $u = \frac{1}{x}, v = \frac{1}{y}, w = \frac{1}{z}$
Then the system of equations

$$-u + 3v + 4w = 30$$
$$3u + 2v - w = 9$$
$$2u - v + 2w = 10$$

The matrix form of the system is

$$\begin{bmatrix} -1 & 3 & 4 \\ 3 & 2 & -1 \\ 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 30 \\ 9 \\ 10 \end{bmatrix}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 3 & 2 & -1 & 9 \\ 2 & -1 & 2 & 10 \end{array}\right]$$

$$R_1 \to (-1)R_1$$
$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 3 & 2 & -1 & 9 \\ 2 & -1 & 2 & 10 \end{array}\right]$$

$$R_2 \to R_2 - 3R_1, R_3 \to R_3 - 2R_1$$
$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 0 & 11 & 11 & 99 \\ 0 & 5 & 10 & 70 \end{array}\right]$$

$$R_2 \to \left(\frac{1}{11}\right) R_2, R_3 \to \left(\frac{1}{5}\right) R_3$$

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 0 & 1 & 1 & 9 \\ 0 & 1 & 2 & 14 \end{array}\right]$$

$$R_3 \to R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 0 & 1 & 1 & 9 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

The corresponding system of equations is

$$u - 3v - 4w = -30$$
$$v + w = 9$$
$$w = 5$$

By doing back substitution we get

$$v + 5 = 9 \Rightarrow v = 4 \Rightarrow y = \frac{1}{4}$$

$$u - 12 - 20 = -30 \Rightarrow u = 2 \Rightarrow x = \frac{1}{2}$$

Hence, $x = \frac{1}{2}, y = \frac{1}{4}, z = \frac{1}{5}$ is required **unique solution** of the system.

Example:

Solve the system of linear equations $x - 2y + z = 1;\ -x + y - z = 0; 2x - y + z = -1$ using Gauss Elimination method.

Solution: The Matrix form of the system of equation

$$\begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Let A= $\begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}\ B = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$$[A|B] = \begin{bmatrix} 1 & -2 & 1 & 1 \\ -1 & 1 & -1 & 0 \\ 2 & -1 & 1 & -1 \end{bmatrix}$$

$$R_2 \to R_1 + R_2 R_3 \to R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 3 & -1 & -3 \end{bmatrix}$$

$$R_2 \to -R_2$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 3 & -1 & -3 \end{bmatrix}$$

$$R_3 \to R_3 - 3R_2$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$-z = 0 => z = 0$$
$$y = -1$$
$$x - 2y + z = 1$$
$$x = -1$$

Hence, the required solution is $x = -1, y = -1\ and\ z = 0$

Example: Solve the system of linear equation using Gauss Elimination method

$$x + y + z = 6; x + 2y + 3z = 14; 2x + 4y + 7z = 30$$

Solution:
$$[A|B] = \begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 1 & 2 & 3 & | & 14 \\ 2 & 4 & 7 & | & 30 \end{bmatrix}$$

$$R_2 \to R_2 - R_1 R_3 \to R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 8 \\ 0 & 2 & 51 & | & 8 \end{bmatrix}$$

$$R_3 \to R_3 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 8 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

$$z = 2$$

$$y + 2z = 8$$

$$\therefore y = 4$$

$$x = 0$$

Therefore, required solution is $x = 0; y = 4; z = 8$

Example: Solve the following system by Gauss elimination Method

$$2x + y + z = 0$$

$$3x + 2y + 3z = 18$$

$$x + 4y + 9z = 16$$

Solution: The corresponding augmented matrix is

$$[A \mid B] = \begin{bmatrix} 2 & 1 & 1 & | & 0 \\ 3 & 2 & 3 & | & 18 \\ 1 & 4 & 9 & | & 16 \end{bmatrix} \qquad R_1 \to \left(\frac{1}{2}\right) R_1$$

$$\sim \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & | & 0 \\ 3 & 2 & 3 & | & 18 \\ 1 & 4 & 9 & | & 16 \end{bmatrix} \qquad R_2 \to R_2 - 3R_1,\ R_3 \to R_3 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & 18 \\ 0 & \frac{7}{2} & \frac{17}{2} & 16 \end{array}\right] \quad R_2 \to 2R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & 18 \\ 0 & \frac{7}{2} & \frac{17}{2} & 16 \end{array}\right] \quad R_3 \to R_3 - \frac{7}{2}R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 3 & 36 \\ 0 & 0 & -2 & -110 \end{array}\right] \quad R_3 \to -\frac{1}{2}R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 3 & 36 \\ 0 & 0 & 1 & 55 \end{array}\right]$$

The resulting system is an upper triangular, which yields,

$$x + \frac{1}{2}y + \frac{1}{2}z = 0$$
$$y + 3z = 36$$
$$z = 55$$

Back substitution gives the solution:

$$z = 55, y = -129, x = 37$$

Example: Solve the system of Linear equations by Gauss elimination method:

$$x - 2y + w = 3$$
$$-x + 2y + z - \frac{1}{2}w = -7$$
$$4x - 8y + 6z + 7w = -3$$

Solution: The corresponding augmented matrix is

$$[A\,|\,B]=\begin{bmatrix} 1 & -2 & 0 & 1 & | & 3 \\ -1 & 2 & 1 & \frac{-1}{2} & | & -7 \\ 4 & -8 & 6 & 7 & | & -3 \end{bmatrix} \quad R_2 \to R_2 + R_1,\ R_3 \to R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 1 & | & 3 \\ 0 & 0 & 1 & \frac{1}{2} & | & -4 \\ 0 & 0 & 6 & 3 & | & -15 \end{bmatrix} \quad R_3 \to R_3 - 6R_2$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 1 & | & 3 \\ 0 & 0 & 1 & \frac{1}{2} & | & -4 \\ 0 & 0 & 0 & 0 & | & 9 \end{bmatrix}$$

The resultant system is,

$$x - 2y + w = 3$$
$$z + \frac{1}{2}w = -4$$
$$0 = 9$$

The last equation, 0 = 9 is never satisfied. Thus, the system has no solution. Hence, the system is inconsistent.

Example: Find the value of λ so that the equations:

$$\begin{aligned} 2x + y + 2z &= 0 \\ x + y + 3z &= 0 \\ 4x + 3y + \lambda z &= 0 \end{aligned} \quad \text{have a nontrivial solution.}$$

Solution: The corresponding

$$[A\,|\,B]=\begin{bmatrix} 2 & 1 & 2 & | & 0 \\ 1 & 1 & 3 & | & 0 \\ 4 & 3 & \lambda & | & 0 \end{bmatrix} \quad R_1 \to \frac{1}{2}R_1$$

$$\sim \begin{bmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 1 & 1 & 3 & | & 0 \\ 4 & 3 & \lambda & | & 0 \end{bmatrix} \quad R_2 \to R_2 - R_1, R_3 \to R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & \frac{1}{2} & 2 & | & 0 \\ 0 & 1 & \lambda - 4 & | & 0 \end{bmatrix} \quad R_2 \to 2R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & 1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 1 & \lambda-4 & 0 \end{array}\right] \quad R_1 \to R_1 - \frac{1}{2}R_2, R_3 \to R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & \lambda-8 & 0 \end{array}\right]$$

Thus, the system to have the nontrivial solution, $\lambda - 8 = 0, \quad \therefore \lambda = 8$

Example: Find the equation of the circle in the form $x^2 + y^2 + 2gx + 2hy + c = 0$ which passes through the points $(6,8),(8,4)$ and $(3,9)$.

Solution: Substituting the values of x and y, we obtain the following in terms of *g, h and c:*

$$12g + 16h + c = -100$$
$$16g + 8h + c = -80$$
$$6g + 18h + c = -90$$

On solving this system by using Gauss elimination method. The corresponding augmented matrix is

$$[A \mid B] = \left[\begin{array}{ccc|c} 12 & 16 & 1 & -100 \\ 16 & 8 & 1 & -80 \\ 6 & 18 & 1 & -90 \end{array}\right] \quad R_1 \to \left(\frac{1}{12}\right)R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & \frac{1}{12} & \frac{-25}{3} \\ 16 & 8 & 1 & -80 \\ 6 & 18 & 1 & -90 \end{array}\right] \quad R_2 \to R_2 - 16R_1,\ R_3 \to R_3 - 6R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & \frac{1}{12} & \frac{-25}{3} \\ 0 & \frac{-40}{3} & \frac{-1}{3} & \frac{160}{3} \\ 0 & 10 & \frac{1}{2} & -40 \end{array}\right] \quad R_2 \to \left(-\frac{3}{40}\right)R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & \frac{1}{12} & \frac{-25}{3} \\ 0 & 1 & \frac{1}{40} & -4 \\ 0 & 10 & \frac{1}{2} & -40 \end{array}\right] \quad R_3 \to R_3 - 10R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & \frac{1}{12} & \frac{-25}{3} \\ 0 & 1 & \frac{1}{40} & -4 \\ 0 & 0 & \frac{1}{4} & 0 \end{array}\right] \quad R_3 \to 4R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & \frac{1}{12} & \frac{-25}{3} \\ 0 & 1 & \frac{1}{40} & -4 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

This gives the following system:

$$g + \frac{4}{3}h + \frac{1}{12}c = \frac{-25}{3}$$
$$h + \frac{1}{40}c = -4$$
$$c = 0$$

Solving, we get $c = 0, h = -4, g = -3$

The equation of a circle is $x^2 + y^2 - 6x - 8y = 0$

Example: Suppose an electrical manufacturing company produces three types of articles line conditioner, servo stabilizer and UPS. The total number of productions is 50 units. The company produce UPS twice the number of line conditioners. Again the total of servo stabilizers and line conditioners is two more than the number of UPS. Find the actual number of each items being produced by the company.

Solution: Let *x, y, z* be the numbers of line conditioners, servo stabilizers, UPS respectively. From the given data, we can construct the following system of equations:

$$x + y + z = 50$$
$$2x - y = 0$$
$$x + y - z = 2$$

The corresponding augmented matrix is

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 2 & 0 & -1 & 0 \\ 1 & 1 & -1 & 2 \end{array}\right] \quad R_2 \to R_2 - 2R_1, R_3 \to R_3 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & -2 & -3 & -100 \\ 0 & 0 & -2 & -48 \end{array}\right] \quad R_2 \to -\frac{1}{2}R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & \frac{3}{2} & 50 \\ 0 & 0 & -2 & -48 \end{array}\right] \quad R_3 \to \left(-\frac{1}{2}\right)R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & \frac{3}{2} & 50 \\ 0 & 0 & 1 & 24 \end{array}\right]$$

The system is

$$x + y + z = 50$$
$$y + \frac{3}{2}z = 50$$
$$z = 24$$

Solution of this system is $z = 24, y = 14, x = 12$

Therefore, the number of line conditioners = 12, servo stabilizer = 14, UPS=24

Exercise:

Solve the following system of equations by using Gauss elimination method.

(1) $x + y + 2z = 9$
$2x + 4y - 3z = 1$
$3x + 6y - 5z = 0$

Ans:
$x = 1, y = 2, z = 3$

(2) $3x + y - 3z = 13$
$2x - 3y + 7z = 5$
$2x + 19y - 47z = 32$

Ans: No solution as the augmented matrix in row-echelon form is

$$\left[\begin{array}{ccc|c} 1 & 1/3 & -1 & 13/3 \\ 0 & 1 & -27/11 & 1 \\ 0 & 0 & 0 & 5 \end{array}\right]$$

(3) $2x + 2y + 2z = 0$
$-2x + 5y + 2z = 1$
$8x + y + 4z = -1$

Ans: Infinitely many solutions. The solution set is $\{(\frac{-3k-1}{7}, \frac{1-4k}{7}, k) / \text{k} \in \text{R}\}$.

(4) $x + y + 2z = 9;$
$2x + 4y - 3z = 1;\ 3x + 6y - 5z = 0$

Answer: $x = 1; y = 2;\ z = 3$

(5) $-x + 3y + 4z = 30;\ 3x + 2y - z = 9;\ 2x - y + 2z = 10$

Answer: $x = 2; y = 4; z = 5$

(6) $x + 2y + z = 5;$

$x = 2, y = 1, z = 1$

$3x - y + z = 6; \quad x + y + 4z = 7$

(7) $x + y + z = 6;$ $\qquad x = 0, y = 4, z = 2$

$x + 2y + 3z = 14;$

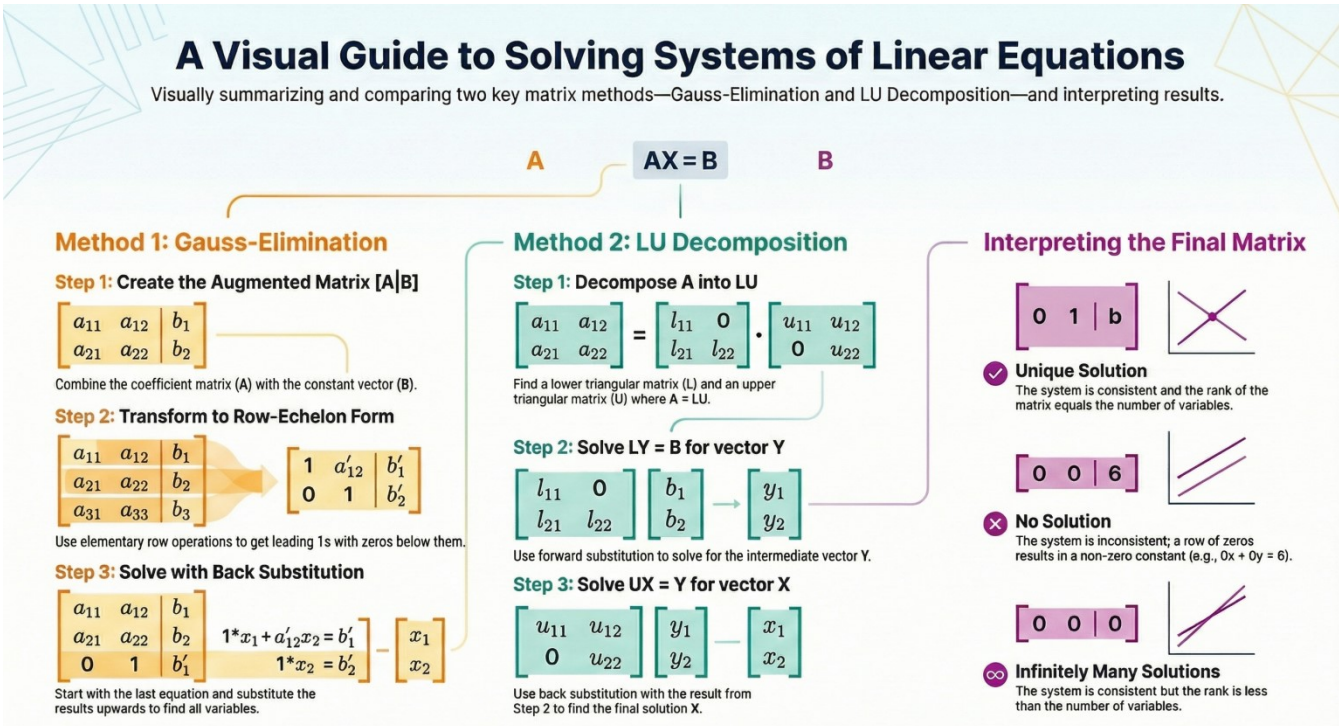
$2x + 4y + 7z = 30.$

(8) Find the values of λ and μ so that the equations:

$$2x + 3y + 5z = 9$$
$$7x + 3y - 2z = 8$$
$$2x + 3y + \lambda z = \mu$$

has (i) unique solution (ii) no solution (iii) infinitely many solutions

(9) Find the quadratic equation $y = ax^2 + bx + c$ that passes through the point (3,18), (2,9) and (-2,13)



LU Decomposition in Linear Algebra:

LU decomposition is a method of decomposing a square matrix A into the product of two matrices: A lower triangular matrix L and an upper triangular matrix U.

This is particularly useful for solving linear systems of equations, computing determinants, and performing matrix inversion efficiently.

If **A** is a matrix, the LU decomposition expresses it as:

Where:

(i)**L** is a lower triangular matrix (all elements above the diagonal are zero).

(ii) **U** is an upper triangular matrix (all elements below the diagonal are zero).

In some cases, a permutation matrix **P** may also be needed to reorder the rows of **A** to ensure stability, giving:

Procedure for LU Decomposition:

Consider the system of linear equations in three variables:

$$a_{11x} + a_{12}y + a_{13}z = b_1; a_{21x} + a_{22}y + a_{23}z = b_2; a_{31x} + a_{32}y + a_{33}z = b_3$$

These can be written in the form of AX =B as:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Here A= $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, X= $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, B = $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Now follow the steps given below to solve the above system of linear equations by LU Decomposition method.

Step-1: Generate a matrix A = LU such that L is the lower triangular matrix with principal diagonal elements being equal to 1 and U is the upper triangular matrix.

That means, L = $\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$ and U = $\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$

Step 2: Now, we can write AX = B as: LUX = B….(1)

Step 3: Let us assume UX = Y….(2)

Where Y = $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

Step 4: From equations (1) and (2), we have; LY = B

On solving this equation, we get y1, y2, y3.

Step 5: Substituting Y in equation (2), we get UX = Y

By solving equation, we get X= (x , y, z)

The above process is also called the Method of Triangularisation.

<u>Conditions for LU Decomposition</u>

- Let A must be a square matrix.

- LU decomposition may fail if **A** is singular or if pivot elements are zero.

Example: Solve the following system using LU Decomposition Method:

$$x + 5y + z = 14, 2x + y + 3z = 13, 3x + y + 4z = 17$$

Solution: We can solve the system using LU Decomposition
Let A = LU and Substitute into AX=B
So LUX =B for X to solve the system…..(1)
Let UX=Y ∴ LY=B and UX=Y
First we solve LY = B and then solve UX=Y for X
We need to find L and U such that A =LU.

$$\begin{bmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

u_{11}=1 , u_{12}=5, u_{13}= 1

$l_{21}u_{11}$ =2 ⇒ $l_{21} = 2$

And $l_{31}u_{11} = 3 \Rightarrow l_{31} = 3$

$l_{21}u_{12} + u_{22}$ =1 ⇒ $u_{22} = -9$ and

$l_{21}u_{13} + u_{23} = 3 \Rightarrow u_{23} = 1$

$l_{31}u_{12} + l_{32}u_{22}$=1 ⇒ $l_{32} = 14/9$

$l_{31}u_{13} + l_{32}u_{23} + u_{33}$=4 ⇒ $u_{33} = -5/9$

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 14/9 & 1 \end{bmatrix} \begin{bmatrix} 1 & 5 & 1 \\ 0 & -9 & 1 \\ 0 & 0 & -5/9 \end{bmatrix}$$

From (i) L<u>UX</u>=B ⇒ LY=B

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 14/9 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \\ 17 \end{bmatrix}$$

y_1= 14 , $2y_2 + y_3$=13 , $3y_1 + \frac{14}{9}y_2 + y_3 = 17$

⇒ y_1= 14 , $y_2 = -15$, $y_3 = -5/3$

Now UX=Y

$$\begin{bmatrix} 1 & 5 & 1 \\ 0 & -9 & 1 \\ 0 & 0 & -5/9 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ -15 \\ -5/3 \end{bmatrix}$$

$\therefore x + 5y + z = 14$,$-9y + z = -15$, z = 3

$\therefore x = 1$,$y = 2$, z = 3

Example-17: Solve the following system using LU Decomposition Method:

$$x + y - z = 4, x - 2y + 3z = -6, 2x + 3y + z = 7$$

Solution: $\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \\ 7 \end{bmatrix}$

AX=B

We can solve the system using LU Decomposition

Let A = LU and Substitute into AX=B

So LUX =B for X to solve the system…..(1)

Let UX=Y ∴ LY=B and UX=Y

First we solve LY = B and then solve UX=Y for X

We need to find L and U such that A =LU.

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

u_{11}=1 , u_{12}=1, u_{13}= -1

$l_{21}u_{11}$ =1 $\Rightarrow l_{21} = 1$ and $l_{31}u_{11} = 2 \Rightarrow l_{31} = 2$

$l_{21}u_{12} + u_{22}$ =-2 $\Rightarrow u_{22} = -1$ and $l_{21}u_{13} + u_{23} = 3 \Rightarrow u_{23} = 2$

$l_{31}u_{12} + l_{32}u_{22}$= 3 $\Rightarrow l_{32} = -1/3$

$l_{31}u_{13} + l_{32}u_{23} + u_{33}$=1 $\Rightarrow u_{33} = 11/3$

A =LU = $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 11/3 \end{bmatrix}$

Now we know that AX=B

For finding solution of equation, we are using

LUX=B ⇒ LY=B

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \\ 17 \end{bmatrix}$$

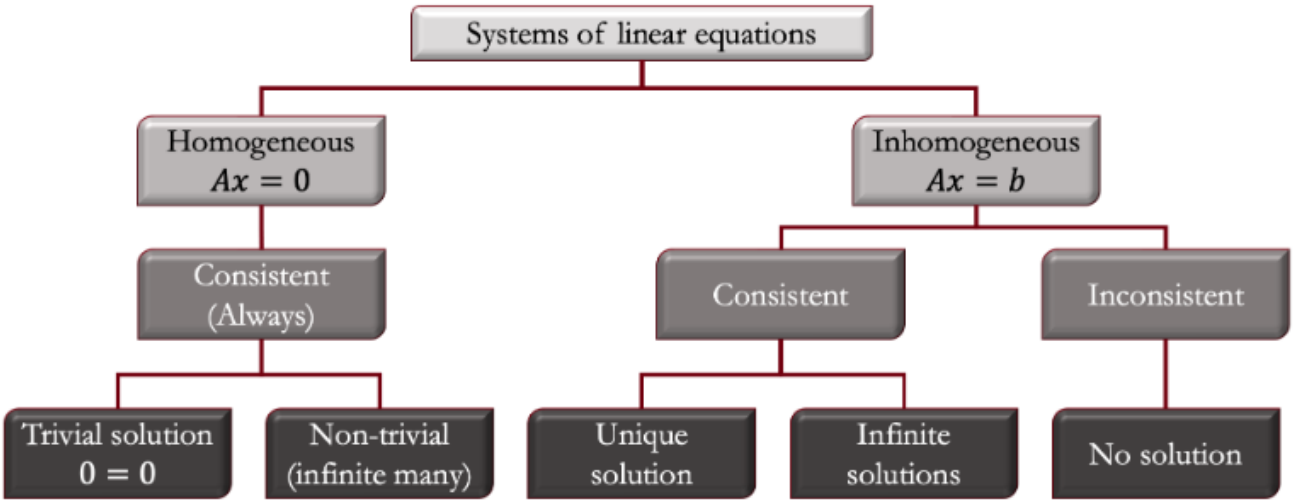
$\Rightarrow y_1 = 4$, $y_2 = -10$, $y_3 = -13/3$

Now UX=Y implies $\begin{bmatrix} 1 & 5 & 1 \\ 0 & -9 & 1 \\ 0 & 0 & -5/9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ -13/3 \end{bmatrix}$

$\therefore x = 1$,$y = 2$, z = -1

Exercise:

1. Solve the system of equations x + y + z = 1, 3x + y – 3z = 5 and x – 2y – 5z = 10 by LU decomposition method. (Ans: (6, -7,2))
2. Solve the below given system of equations by LU decomposition.
   x + y + z = 1, 4x + 3y – z = 6, 3x + 5y + 3z = 4
3. Find the solution of the system of equations by LU decomposition.
   x + 2y + 3z = 9, 4x + 5y + 6z = 24, 3x + y – 2z = 4



Classification of types of Solution of Systems of Linear equations

1.5 HOMOGENEOUS EQUATIONS

A system of linear equations in terms of $x_1, x_2, x_3, \ldots x_n$ having the matrix form AX= O, where A is $m \times n$ coefficient matrix, X is $n \times 1$ column matrix, O is a $m \times 1$ zero column matrix is called a system of homogeneous equations

e.g.: $x + y + z = 0$; $x + 2y - z = 0$; $x + 3y + 2z = 0$

e.g.: $x + y = 0$; $x + 2y = 0$

Homogeneous equations are never inconsistent. They always have the solution "all variables = 0". The solution (0, 0, …, 0) is often called the **trivial solution**. Any other solution is called **nontrivial solution.**

Example: Solve the following system:

$$4x + 3y - z = 0$$
$$3x + 4y + z = 0$$
$$5x + y - 4z = 0$$

Solution: The matrix form of the system is

$$\begin{bmatrix} 4 & 3 & -1 \\ 3 & 4 & 1 \\ 5 & 1 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix is

$$[A|B]$$
$$\left[\begin{array}{ccc|c} 4 & 3 & -1 & 0 \\ 3 & 4 & 1 & 0 \\ 5 & 1 & -4 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 4 & 3 & -1 & 0 \\ 3 & 4 & 1 & 0 \\ 5 & 1 & -4 & 0 \end{array}\right] R_1 \to \frac{1}{4}R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3/4 & -1/4 & 0 \\ 3 & 4 & 1 & 0 \\ 5 & 1 & -4 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 3R_1, \\ R_3 \to R_3 - 5R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 3/4 & -1/4 & 0 \\ 0 & 7/4 & 7/4 & 0 \\ 0 & -11/4 & -11/4 & 0 \end{array}\right] R_2 \to \frac{4}{7}R_2$$

$=$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x - z = 0,\ y + z = 0,\ 0 = 0.$

The last equation does not give any information about the equations.

Let. $z = k \Rightarrow y = -k \text{ and } x = k$

$\therefore \text{the solution set is } \{(k, -k, k) / k \in R\}$

Exercise: Solve the following system of equations.

(1)

$x + y - z + w = 0$

$x - y + 2z - w = 0$

Ans: Infinitely many solutions.

The solution set is

Example: Solve the following system

$-2x + 2y - 3z = 0$

$2x + y - 6z = 0$

$-x - 2y + 2z = 0$

$3x + y + 4z = 0$

Solution:

$$\left[\begin{array}{ccc|c} -2 & 2 & -3 & 0 \\ 2 & 1 & -6 & 0 \\ -1 & -2 & 2 & 0 \\ 3 & 1 & 4 & 0 \end{array}\right] R_1 \to -\frac{1}{2}R_1$$

$=$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 0 \\ 2 & 1 & -6 & 0 \\ -1 & -2 & 2 & 0 \\ 3 & 1 & 4 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 2R_1, \\ R_3 \to R_3 + R_1 \\ R_4 \to R_4 - 3R_1 \end{array}$$

$=$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 0 \\ 0 & 3 & -9 & 0 \\ 0 & -3 & 7/2 & 0 \\ 0 & 4 & -1/2 & 0 \end{array}\right] R_2 \to \frac{1}{3}R_2$$

$=$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -3 & 7/2 & 0 \\ 0 & 4 & -1/2 & 0 \end{array}\right] \begin{array}{l} R_1 \to R_1 + R_2, \\ R_3 \to R_3 + 3R_2 \\ R_4 \to R_4 - 4R_2 \end{array}$$

$=$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3/2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -11/2 & 0 \\ 0 & 0 & 23/2 & 0 \end{array}\right] R_3 \to -\frac{2}{11}R_3$$

$=$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3/2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 23/2 & 0 \end{array}\right] \begin{array}{l} R_1 \to R_1 + 3/2R_3, \\ R_2 \to R_2 + 3R_3 \\ R_4 \to R_4 - 23/2R_3 \end{array}$$

$3x + y \quad + w = 0$ (2) $2x + y + 3z = 0$ $x + 2y \quad = 0$ $y + z = 0$	$\{(t/4, -7t/4, t)/\ t \in R\}$. Ans: Trivial solution $x = 0, y = 0, z = 0$	$= \begin{bmatrix} 1 & 0 & 0 & \vert & 0 \\ 0 & 1 & 0 & \vert & 0 \\ 0 & 0 & 1 & \vert & 0 \\ 0 & 0 & 0 & \vert & 0 \end{bmatrix}$ The required solution is $x = 0, y = 0, z = 0$ which is trivial solution.

1.6 Rank of a Matrix

The positive integer r is said to be a rank of a matrix A if it possesses the following properties:

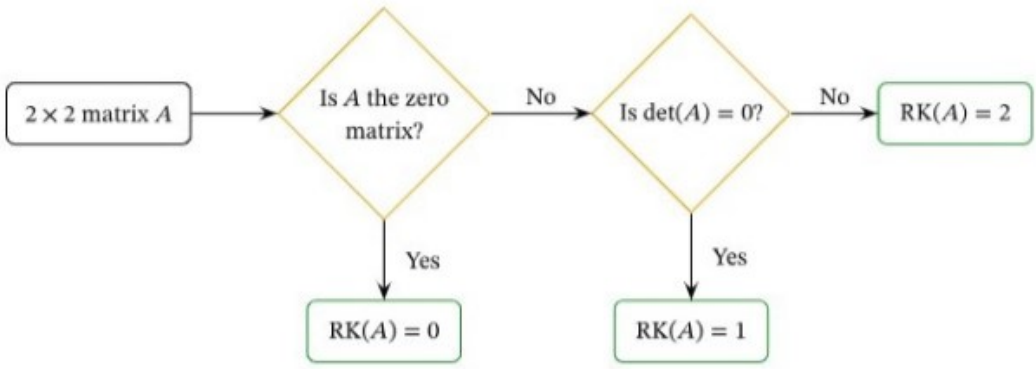
(i)There is at least one minor of order r which is non-zero.

(ii) Every minor of order greater than r is zero.

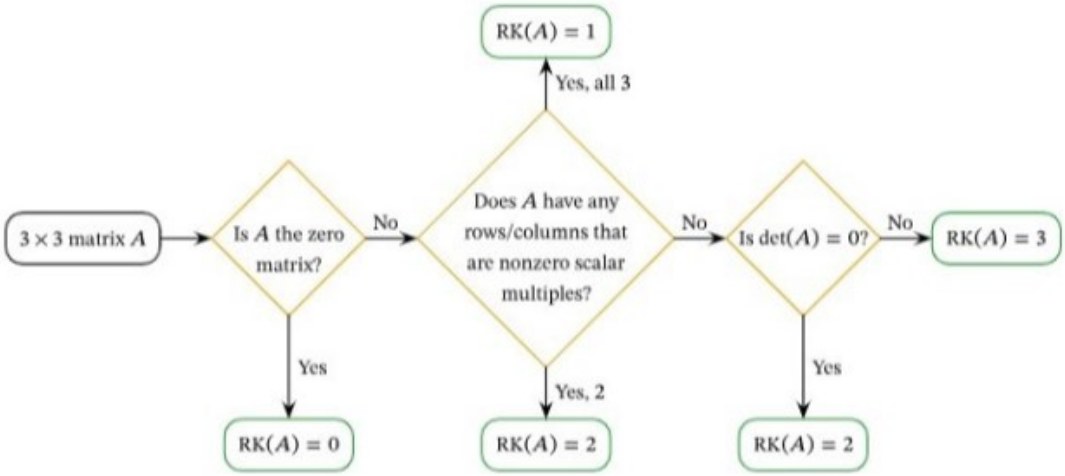
Remarks:

1. Rank of matrix A is denoted by $\rho(A)$
2. The rank of matrix remains unchanged by elementary transformation
3. $\rho(A^T) = \rho(A)$
4. The rank of the product of two matrices always less than or equal to the rank of either matrix (i.e., $\rho(AB) \leq \rho(A)$ or $\rho(AB) \leq \rho(B)$).

- The rank of a 2×2 matrix A can be found by the following process:



- The rank of a 3×3 matrix A can be found by the following process:



Flowcharts to find rank for 2 X2 and 3 X3 matrices

1.6.1 Methods for finding Rank of a Matrix:

❖ **Method-1: Rank of a Matrix by Determinant Matrix**

Consider a square matrix A of order r.

- **Step-1:** Find the determinant of A. If $det(A) \neq 0$ then $\rho(A) = r$. Otherwise $\rho(A) < r$.
- **Step-2:** Find the all-possible minors of order $r - 1$. If any one of them is non-zero then order is $r - 1$, otherwise $\rho(A) < r - 1$.
- **Step-3:** By continuing this process upto the non-zero determinant.

Example: Find the rank the following matrices by determinant method:

a. $\boldsymbol{A} = \begin{bmatrix} \mathbf{2} & \mathbf{3} & \mathbf{4} \\ \mathbf{4} & \mathbf{3} & \mathbf{1} \\ \mathbf{1} & \mathbf{2} & \mathbf{4} \end{bmatrix}$

Solution: Given, $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ then $det(A) \neq 0$. Hence, the $\rho(A) = 3$

b. $\boldsymbol{A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}}$

Solution: Given, $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$ then $det(A) = 0$. Hence, the rank of A is less than 3.

Now, minor of $1 = \begin{vmatrix} 3 & 4 \\ 5 & 7 \end{vmatrix} = 21 - 20 = 1 \neq 0$. Hence, $\rho(A) = 2$.

c. $\boldsymbol{A = \begin{bmatrix} 4 & 2 & 3 \\ 8 & 4 & 6 \\ -2 & -1 & -\frac{3}{2} \end{bmatrix}}$

Solution: Given, $A = \begin{bmatrix} 4 & 2 & 3 \\ 8 & 4 & 6 \\ -2 & -1 & -\frac{3}{2} \end{bmatrix}$ then $det(A) = 0$. Hence $\rho(A) < 3$.

Consider all the minors of order 2, i.e.,

$$\begin{vmatrix} 4 & 2 \\ 8 & 4 \end{vmatrix} = 0, \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 0, \begin{vmatrix} 4 & 3 \\ 8 & 6 \end{vmatrix} = 0, \begin{vmatrix} 4 & 2 \\ -2 & -1 \end{vmatrix} = 0, \begin{vmatrix} 2 & 3 \\ -1 & -\frac{3}{2} \end{vmatrix} = 0, \begin{vmatrix} 4 & 3 \\ -2 & -\frac{3}{2} \end{vmatrix} = 0$$

Here, all the minors of order 2 are zero. There rank is less than 2. Hence, $\rho(A) = 1$.

d. $\boldsymbol{A = \begin{bmatrix} 1 & 2 & -1 & -4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & -7 \end{bmatrix}}$

Solution: Here, the order of matrix A is 3×4. Hence the rank of A is maximum 3 as we can find the square matrix of order 3. Therefore, consider all the minors of order 3, i.e.,

$$\begin{vmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ -1 & -2 & 6 \end{vmatrix} = 0, \begin{vmatrix} 2 & -1 & -4 \\ 4 & 3 & 5 \\ -2 & 6 & -7 \end{vmatrix} = 0, \begin{vmatrix} 1 & 2 & -4 \\ 2 & 4 & 5 \\ -1 & -2 & -7 \end{vmatrix} = 0, \begin{vmatrix} 1 & -1 & -4 \\ 2 & 3 & 5 \\ -1 & 6 & -7 \end{vmatrix} = -120$$

Here, one minor of rank 3 is not equal to zero. Hence, $\rho(A) = 3$.

Example: Find the value of p for which the following matrix A will be of
(i) rank one (ii) rank two (iii) rank three

$$A = \begin{bmatrix} 3 & p & p \\ p & 3 & p \\ p & p & 3 \end{bmatrix}$$

Solution:

$$\det(A)=\begin{vmatrix}3 & p & p\\ p & 3 & p\\ p & p & 3\end{vmatrix}$$

$$=3(9-p^2)-p(3p-p^2)+p(p^2-3p)$$

$$=(3-p)(9+3p-p^2-p^2)$$

$$=(p-3)(2p^2-3p-9)$$

$$=(p-3)(p-3)(2p+3)$$

$$=(p-3)^2(2p+3)$$

(i) *If* $p=3$

$$A=\begin{bmatrix}3 & 3 & 3\\ 3 & 3 & 3\\ 3 & 3 & 3\end{bmatrix}$$

$$\det(A)=0$$

And all the minors of order 2 are zero.
Hence, $\rho(A)=1$
Rank of A will be 1 if $p = 3$

(ii) Rank of A will be 2 if *det(A)=0*, but $p\neq 3$

$$(p-3)^2(2p+3)=0 \text{ but } p\neq 3$$

$$2p+3=0$$

$$p=\frac{-3}{2}$$

(iii) Rank of A will be 3 if $\det(A)\neq 0$

$$(p-3)^2(2p+3)\neq 0$$

$$p\neq 3,\ p\neq\frac{-3}{2}$$

Example: For what value of x, will the matrix $A=\begin{bmatrix}3-x & 2 & 2\\ 1 & 4-x & 0\\ -2 & -4 & 1-x\end{bmatrix}$ be of rank

(i) Equal to 3 (ii) less than 3

$$\det(A)=\begin{vmatrix}3-x & 2 & 2\\ 1 & 4-x & 0\\ -2 & -4 & 1-x\end{vmatrix}$$

$$=(3-x)[(4-x)(1-x)-0]-2(1-x)+2(-4+8-2x)$$
$$=(3-x)(4-x)(1-x)+2(3-x)$$
$$=(3-x)(x^2-5x+6)$$
$$=(3-x)(x-3)(x-2)$$
$$=-(x-3)^2(x-2)$$

Solution:

$$(i)\ \rho(A)=3 \quad if \quad \det(A)\neq 0$$
$$(x-3)^2(x-2)\neq 0$$
$$x\neq 2,3$$
$$(ii)\ \rho(A)<3 \quad if \quad \det(A)=0$$
$$(x-3)^2(x-2)=0$$
$$x=2,3$$

Example: Find the possible values of p, for which the following matrix A will have

(i) Rank 1 (ii) rank 2 (iii) rank 3

$$A=\begin{bmatrix}p & p & 2\\ 2 & p & p\\ p & 2 & p\end{bmatrix}$$

Ans: (i) p=2; (ii) p=-2; (iii) $p\neq -1, p\neq 2$

Example: Find the value of λ for which rank of the matrix $A=\begin{bmatrix}3 & 1 & 2\\ -1 & 4 & 5\\ 7 & 2 & \lambda\end{bmatrix}$

(i) Is less than 3 (ii) equal to 3

❖ **<u>Method-2: Rank of a Matrix by Row Echelon Form</u>**

The Rank of a Matrix in Row Echelon Form is equal to the number of non-zero rows of the matrix.

e.g. A=$\begin{vmatrix}1 & 3 & -1\\ 0 & 1 & 4\\ 0 & 0 & 0\end{vmatrix}$; the matrix A is in Row Echelon form with two non-zero rows. Hence, rank of matrix A is 2.

Example: Find the rank the following matrices by reducing to Row Echelon Form:

(a) $A=\begin{bmatrix}\mathbf{5} & \mathbf{3} & \mathbf{14} & \mathbf{4}\\ \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{1}\\ \mathbf{1} & \mathbf{-1} & \mathbf{2} & \mathbf{0}\end{bmatrix}$ Solution: Given $A=\begin{bmatrix}5 & 3 & 14 & 4\\ 0 & 1 & 2 & 1\\ 1 & -1 & 2 & 0\end{bmatrix}$ By applying row-operations $R_{13}\sim\begin{bmatrix}1 & -1 & 2 & 0\\ 0 & 1 & 2 & 1\\ 5 & 3 & 14 & 4\end{bmatrix}$ $R_3-5R_1\sim\begin{bmatrix}1 & -1 & 2 & 0\\ 0 & 1 & 2 & 1\\ 0 & 8 & 4 & 4\end{bmatrix}$ $R_3-8R_2\sim\begin{bmatrix}1 & -1 & 2 & 0\\ 0 & 1 & 2 & 1\\ 0 & 0 & -12 & -4\end{bmatrix}$ $\left(-\frac{1}{12}\right)R_3\sim\begin{bmatrix}1 & -1 & 2 & 0\\ 0 & 1 & 2 & 1\\ 0 & 0 & 1 & \frac{1}{3}\end{bmatrix}$ The equivalent matrix is in Row-Echelon Form. Number of non-zero rows $=3$ .Hence, $\rho(A)=3$	**(b)** $A=\begin{bmatrix}\mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{-1}\\ \mathbf{-2} & \mathbf{-1} & \mathbf{-3} & \mathbf{-1}\\ \mathbf{1} & \mathbf{0} & \mathbf{1} & \mathbf{1}\\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{-1}\end{bmatrix}$ Solution: Given $A=\begin{bmatrix}1 & 2 & 3 & -1\\ -2 & -1 & -3 & -1\\ 1 & 0 & 1 & 1\\ 0 & 1 & 1 & -1\end{bmatrix}$ By applying row-operations $R_2+2R_1, R_3-R_1\sim\begin{bmatrix}1 & 2 & 3 & -1\\ 0 & 3 & 3 & -3\\ 0 & -2 & -2 & 2\\ 0 & 1 & 1 & -1\end{bmatrix}$ $R_{24}\sim\begin{bmatrix}1 & 2 & 3 & -1\\ 0 & 1 & 1 & -1\\ 0 & -2 & -2 & 2\\ 0 & 3 & 3 & -3\end{bmatrix}$ $R_3+2R_2, R_4-3R_2\sim\begin{bmatrix}1 & 2 & 3 & -1\\ 0 & 1 & 1 & -1\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$ The equivalent matrix is in Row-Echelon Form. Number of non-zero rows $=2$.Hence, $\rho(A)=2$

Exercise:

(1) Find the ranks of A, B, AB and verify $\rho(AB)\le\rho(A)$ or $\rho(AB)\le\rho(B)$) where $A=\begin{bmatrix}2 & 4 & 1\\ 3 & 6 & 2\\ 4 & 8 & 3\end{bmatrix}$,

$B=\begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 2\\ 4 & 3 & 5\end{bmatrix}$

(2) Find the rank the following matrices by reducing to Row Echelon Form:

(i) $A=\begin{bmatrix}1 & 1 & -1 & 1\\ 1 & -1 & 2 & -1\\ 3 & 1 & 0 & 1\end{bmatrix}$, (ii) $A=\begin{bmatrix}3 & -2 & 0 & -1 & -7\\ 0 & 2 & 2 & 1 & -5\\ 1 & -2 & -3 & -2 & 1\\ 0 & 1 & 2 & 1 & 6\end{bmatrix}$, (iii) $\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$, (iv) $\begin{bmatrix}1 & 1 & -1 & 1\\ 1 & -1 & 2 & -1\\ 3 & 1 & 0 & 1\end{bmatrix}$

(v) $\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 3 & 4 & 5\end{bmatrix}$

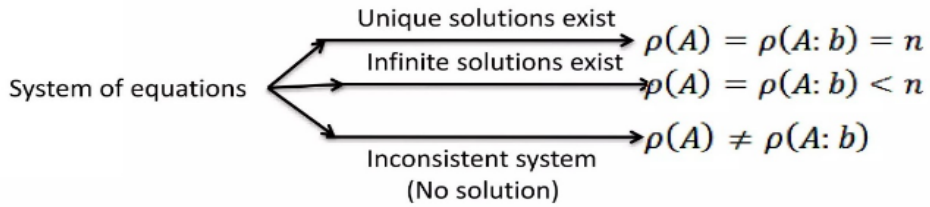
Remarks:

(1) If $\rho(A) \neq \rho(A|B)$ then the system is inconsistent.

(2) If $\rho(A) = \rho(A|B)$ then the system is consistent.

(3) If $\rho(A) < n$ then there are infinitely many solutions (n is the number of unknowns)

(4) If $\rho(A) = n$ then there is a unique solution.



Schematic diagram for types of solutions

Example: What conditions must b_1, b_2, b_3 satisfy for the system to be $x_1+2x_2+3x_3=b_1, 2x_1+5x_2+3x_3=b_2, x_1+8x_3=b_3$ be consistent?

Solution:

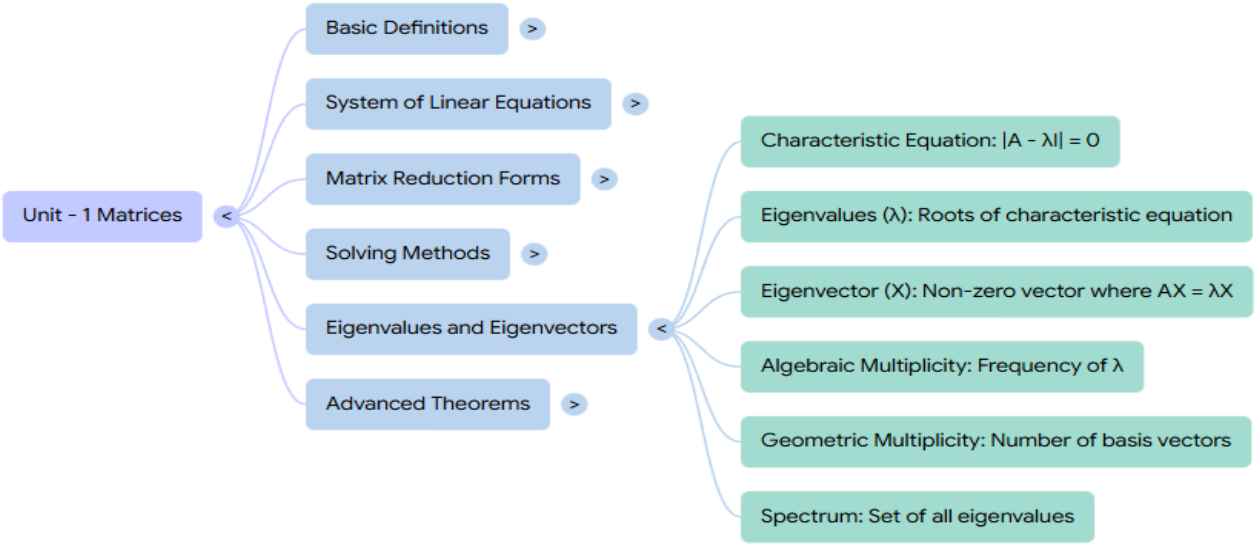
$$[A \mid B]=\left[\begin{array}{ccc|c}1 & 2 & 3 & b_1\\ 2 & 5 & 3 & b_2\\ 1 & 0 & 8 & b_3\end{array}\right] \qquad R_2 \to R_2-2R_1,\ R_3 \to R_3-R_1$$

$$\sim \left[\begin{array}{ccc|c}1 & 2 & 3 & b_1\\ 0 & 1 & -3 & b_2-2b_1\\ 0 & 2 & 5 & b_3-b_1\end{array}\right] \qquad R_3 \to R_3+2R_2$$

$$\sim \left[\begin{array}{ccc|c}1 & 2 & 3 & b_1\\ 0 & 1 & -3 & b_2-2b_1\\ 0 & 0 & -1 & b_3+2b_2-5b_1\end{array}\right] \qquad R_3 \to (-1)R_3$$

$$\sim \left[\begin{array}{ccc|c}1 & 2 & 3 & b_1\\ 0 & 1 & -3 & b_2-2b_1\\ 0 & 0 & 1 & -b_3-2b_2+5b_1\end{array}\right]$$

Since, rank (A) = rank (A | B) = 3, the solution is unique for every choice of the values of b_1, b_2, b_3.



Schematic diagram for Eigen values and Eigen vectors concepts

1.7 Eigen values and Eigen vectors:

Let A be $n \times n$ matrix, then there exists a real number λ and a nonzero vector X such that

$$AX = \lambda X$$

then, λ is called as the eigen value or characteristic value or latent root or proper roots of the matrix$A$, and X is called as eigen vector or characteristic vector or real vector corresponding to eigen value λ of the matrix A.

As, $AX = \lambda X \Rightarrow AX - \lambda\, I_n X = 0 \Rightarrow (A - \lambda\, I_n)X = 0$

Which is a homogeneous system, and has a nontrivial solution if $|A - \lambda\, I_n| = 0$

Remarks:

1. An Eigen vector is never the zero vector.
2. The matrix $[A - \lambda I_n]$ is known as the **characteristic matrix** of A.
3. The determinant of $(A - \lambda I_n)$ after expansion gives the polynomial in λ, it is known as the **characteristic polynomial** of the matrix A of order $n \times n$ and is of degree n.
4. $|A - \lambda I_n|$=0 is called the **characteristic equation** of matrix A.
5. The root of the characteristic equation is known as **characteristic value** or **eigenvalues** of the matrix.
6. The set of all characteristic roots (Eigen values) of the matrix A is called the **spectrum of A.**
7. Let A be $n \times n$ matrix and λ be an eigen value for A. Then the set$E_\lambda = \{X / AX = \lambda X\}$ is called the **Eigen space of λ.**

Results:

1. The Eigen values of a diagonal matrix are its diagonal elements.
2. The sum of Eigen values of an $n \times n$ matrix is its trace and their product is $|A|$.
3. For the upper triangular (lower triangular) $n \times n$ matrix A, the Eigen values are its diagonal elements.

Example:
If $A = \begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix}$, find the Eigen values for the given matrices:
(i) A, (ii) A^T, (iii) A^{-1}, (iv) $4A^{-1}$, (v) A^2, (vi) $A^2 - 2A + I$,
(vii) $A^3 + 2I$

Solution: Given, $A = \begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix}$

The characteristic equation of matrix A is

$$|A - \lambda I_2| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 4 \\ 3 & 2-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)(2-\lambda) - 12 = 0 \Rightarrow \lambda^2 - 3\lambda - 10 = 0 \Rightarrow (\lambda - 5)(\lambda + 2) = 0$$

$$\therefore \lambda = 5 \text{ or } \lambda = -2$$

Eigenvalues of $A = \lambda$	$5, -2$
Eigenvalues of $A^T = \lambda^T$	$5, -2$
Eigenvalues of $A^{-1} = \lambda^{-1}$	$\frac{1}{5}, -\frac{1}{2}$
Eigenvalues of $4A^{-1} = 4\lambda^{-1}$	$\frac{4}{5}, -2$
Eigenvalues of $A^2 = \lambda^2$	$25, 4$
Eigenvalues of $A^2 - 2A + I = \lambda^2 - 2\lambda + 1$	$16, 9$
Eigenvalues of $A^3 + 2I = \lambda^3 + 2$	$127, -6$

Example: Find the Eigen values of $A = \begin{vmatrix} 3 & 2 \\ 3 & 8 \end{vmatrix}$

Solution: Given $A = \begin{vmatrix} 3 & 2 \\ 3 & 8 \end{vmatrix}$, then the characteristic equation of matrix A is

$$|A - \lambda I_2| = 0$$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 2 \\ 3 & 8-\lambda \end{vmatrix} = 0 \Rightarrow (3-\lambda)(8-\lambda) - 6 = 0 \Rightarrow \lambda^2 - 11\lambda + 18 = 0 \Rightarrow (\lambda - 9)(\lambda - 2) = 0$$

$$\therefore \lambda_1 = 9 \text{ or } \lambda_2 = 2$$

Example: If $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ then find the eigen values of A^2 and A^{-1}.

Solution: The eigen values are diagonal elements are 1,2 as A is a diagonal matrix.

Therefore, eigen values of A^2 are $(1)^2, (2)^2$ *implies* $1, 4$

and eigen values of A^{-1} are $\frac{1}{1}, \frac{1}{2}$.

Example: If $A=\begin{bmatrix}1 & 2 & -3\\ 0 & 2 & 3\\ 0 & 0 & 2\end{bmatrix}$ then find the eigen values of A^T and $5A$.

Solution: Since A is an upper triangular matrix, the eigen values of A are 1,2,2.
Therefore, the eigen values of A^T are 1,2,2
The eigen values of $5A$ are 5,10,10

Types of Eigen Values

- Eigenvalues are non-repeated, whether matrix is symmetric or non-symmetric
- Eigenvalues are repeated and the matrix is non-symmetric
- Eigenvalues are repeated and the matrix is symmetric

Example: Find the eigen values and eigen vector of the matrix A= $\begin{bmatrix}-2 & -8 & -12\\ 1 & 4 & 4\\ 0 & 0 & 1\end{bmatrix}$

Solution:
The characteristic equation is $|A-\lambda I_n|=0$

$$\begin{vmatrix}-2-\lambda & -8 & -13\\ 1 & 4-\lambda & 4\\ 0 & 0 & 1-\lambda\end{vmatrix}=0$$

$\therefore \lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$

$S_1 = tr(A) = -2+4+1 = 3$

$S_2 =$ Sum of minors of diagonal entries

$$=\begin{vmatrix}4 & 4\\ 0 & 1\end{vmatrix}+\begin{vmatrix}-2 & -12\\ 0 & 1\end{vmatrix}+\begin{vmatrix}1 & 4\\ 0 & 0\end{vmatrix}=4-2+0=2$$

$|A| = -2(4)+8(1)-12(0) = -8+8 = 0$

$\therefore$ characteristic equation is

$\lambda^3 - 3\lambda^2 + 2\lambda = 0$

$\Rightarrow \lambda(\lambda^2 - 3\lambda + 2) = 0$

$\Rightarrow \lambda = 0$ or $\lambda^2 - 3\lambda + 2 = 0$

$\Rightarrow \lambda = 0$ or $(\lambda-2)(\lambda-1) = 0$

$\Rightarrow \lambda = 0$ or $\lambda = 1$ or $\lambda = 2$

Here, one can observe that all eigenvalues are non-repeated and matrix is non-symmetric.

When $\lambda_1 = 0$

$$=\left[\begin{array}{ccc|c}1 & 8/3 & 4 & 0\\ 1 & 3 & 4 & 0\\ 0 & 0 & 0 & 0\end{array}\right] R_2 \to R_2 - R_1$$

$$=\left[\begin{array}{ccc|c}1 & 8/3 & 4 & 0\\ 0 & 1/3 & 0 & 0\\ 0 & 0 & 0 & 0\end{array}\right] R_2 \to 3R_2$$

$$=\left[\begin{array}{ccc|c}1 & 8/3 & 4 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0\end{array}\right] R_1 \to R_1 - 8/3R_2$$

$$=\left[\begin{array}{ccc|c}1 & 0 & 4 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0\end{array}\right]$$

We suppose $z = k, y = 0, x + 4z = 0$

$\therefore z = k, y = 0, x = -4z = -4k$

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-4k\\ 0\\ k\end{bmatrix}=k\begin{bmatrix}-4\\ 0\\ 1\end{bmatrix}$$

Therefore, eigen vector space for $\lambda_1 = 0$

$$[A-\lambda I \mid O]=\left[\begin{array}{ccc|c}-2 & -8 & -12 & 0\\ 1 & 4 & 4 & 0\\ 0 & 0 & 1 & 0\end{array}\right] R_1 \to -1/2R_1$$

$$=\left[\begin{array}{ccc|c}1 & 4 & 6 & 0\\ 1 & 4 & 4 & 0\\ 0 & 0 & 1 & 0\end{array}\right] R_2 \to R_2 - R_1$$

$$=\left[\begin{array}{ccc|c}1 & 4 & 6 & 0\\ 0 & 0 & -2 & 0\\ 0 & 0 & 1 & 0\end{array}\right]$$

Therefore, we suppose
$x+4y+6z=0, -2z=0, y=k$
$\therefore z=0, y=k, x=-4k$
Therefore, eigen vector space is

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-4k\\ k\\ 0\end{bmatrix}=k\begin{bmatrix}-4\\ 1\\ 0\end{bmatrix}$$

Therefore, eigen vector space for $\lambda_1=0$ is

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-4\\ 1\\ 0\end{bmatrix}$$

When $\lambda_2=1$

$$[A-\lambda I \mid O]=\left[\begin{array}{ccc|c}-2-1 & -8 & -12 & 0\\ 1 & 4-1 & 4 & 0\\ 0 & 0 & 1-1 & 0\end{array}\right]$$

$$=\left[\begin{array}{ccc|c}-3 & -8 & -12 & 0\\ 1 & 3 & 4 & 0\\ 0 & 0 & 0 & 0\end{array}\right] R_1 \to -1/3R_1$$

(Continue on previous page …)

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-4\\ 0\\ 1\end{bmatrix}$$

When $\lambda_3=2$

$$[A-\lambda I \mid O]=\left[\begin{array}{ccc|c}-2-2 & -8 & -12 & 0\\ 1 & 4-2 & 4 & 0\\ 0 & 0 & 1-2 & 0\end{array}\right]$$

$$=\left[\begin{array}{ccc|c}-4 & -8 & -12 & 0\\ 1 & 2 & 4 & 0\\ 0 & 0 & -1 & 0\end{array}\right] R_1 \to -1/4R_1$$

$$=\left[\begin{array}{ccc|c}1 & 2 & 3 & 0\\ 1 & 2 & 4 & 0\\ 0 & 0 & -1 & 0\end{array}\right] R_2 \to R_2 - R_1$$

$$=\left[\begin{array}{ccc|c}1 & 2 & 3 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & -1 & 0\end{array}\right] R_3 \to -R_3$$

$$=\left[\begin{array}{ccc|c}1 & 2 & 3 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0\end{array}\right] \begin{array}{l}R_1 \to R_1 - 3R_3\\ R_2 \to R_2 - R_3\end{array}$$

$$=\left[\begin{array}{ccc|c}1 & 2 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{array}\right]$$

We suppose $z=0, y=k, x+2y=0$
$\therefore z=0, y=k, x=-2z=-2k$

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-2k\\ k\\ 0\end{bmatrix}=k\begin{bmatrix}-2\\ 1\\ 0\end{bmatrix}$$

Therefore, Eigen vector space for $\lambda_3=2$

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-2\\ 1\\ 0\end{bmatrix}$$

1.8 Algebraic multiplicity and Geometric multiplicity

Let A be $n \times n$ matrix and λ be an eigen value for A. If λ occurs $(k \geq 1)$ times then k is called the **Algebraic multiplicity** of λ, and the number of basis vectors is called **Geometric multiplicity**.

Example: Find Eigen values and Eigen vectors of the matrix. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. Also determine algebraic and geometric multiplicity.

Solution: The characteristic equation is $|A - \lambda I_n| = 0$.

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{vmatrix}$$

$$= (-2-\lambda)[(1-\lambda)(-\lambda)-(-2)(-6)] - 2[2(-\lambda)-(-1)(-6)] - 3[2(-2)-(-1)(1-\lambda)]$$

$$= (-2-\lambda)[-\lambda+\lambda^2-2] - 2[-2\lambda-6] - 3[-4+1-\lambda]$$

$$= -\lambda^3 - \lambda^2 + 21\lambda + 45$$

$$= -(\lambda^3 + \lambda^2 - 21\lambda - 45)$$

$$\therefore -(\lambda^3 + \lambda^2 - 21\lambda - 45) = 0$$

$$\therefore \lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\therefore (\lambda+3)(\lambda^2 - 2\lambda - 15) = 0$$

$$\Rightarrow (\lambda+3)(\lambda+3)(\lambda-5) = 0$$

$$\Rightarrow \lambda_1 = 5, \lambda_2 = -3, \lambda_3 = -3$$

Algebraic Multiplicity of $\lambda = -3$ is 2 and of $\lambda = 5$ is 1.
We solve the following homogeneous system:

$$\therefore [A - \lambda I]X = \begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Case I: When $\lambda_1 = 5$

$$\therefore [A - \lambda I | O] = \begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} -7 & 2 & -3 & 0 \\ 2 & -4 & -6 & 0 \\ -1 & 2 & -5 & 0 \end{array}\right] R_1 \leftrightarrow R_3$$

$$= \left[\begin{array}{ccc|c} -1 & -2 & -5 & 0 \\ 2 & -4 & -6 & 0 \\ -7 & 2 & -3 & 0 \end{array}\right] R_1 \to -R_1$$

=

Case II: When $\lambda_2 = -3, \lambda_3 = -3$

$$\therefore [A - \lambda I | O] = \begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 4 & -6 & 0 \\ -1 & -2 & 3 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 + R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

which is in Row-Echelon from.
We suppose

$\left[\begin{array}{ccc\|c} 1 & 2 & 5 & 0 \\ 2 & -4 & -6 & 0 \\ -7 & 2 & -3 & 0 \end{array}\right]$ $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 + 7R_1$ $=$ $\left[\begin{array}{ccc\|c} 1 & 2 & 5 & 0 \\ 0 & -8 & -16 & 0 \\ 0 & 16 & 32 & 0 \end{array}\right]$ $R_2 \to -1/8R_2$ $=$ $\left[\begin{array}{ccc\|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 16 & 32 & 0 \end{array}\right]$ $R_1 \to R_1 - 2R_2$, $R_3 \to R_3 - 16R_2$ $=$ $\left[\begin{array}{ccc\|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ which is in Row-Echelon from. We suppose $x_3 = k, x_2 + 2x_3 = 0 \Rightarrow x_2 = -2k,$ $x_1 + x_3 = 0 \Rightarrow x_1 = -k$ Therefore, Eigen space is for $\lambda_1 = 5$ is $\{k(-1,-2,1) / k \in R\}$	$x_2 = k_1, x_3 = k_2, x_1 + 2x_2 - 3x_3 = 0 \Rightarrow x_1 = -2k_1 + 3k_2$ Therefore, Eigen space is for $\lambda_2 = -3, \lambda_3 = -3$ is $\{k_1(-2,1,0) + k_2(3,0,1) / k_1, k_2 \in R\}$ Hence, Geometric multiplicity of $\lambda_2 = -3$ is 2 and of $\lambda = 5$ is 1.

Example: Find Eigen values and Eigen vectors of the matrix. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Also determine algebraic and geometric multiplicity.

<u>Solution</u>: The characteristic equation is $|A - \lambda I_n| = 0$.

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix}$$

$= -\lambda(\lambda^2 - 1) - 1(-\lambda - 1) + 1(1 + \lambda)$

$= -\lambda^3 + 3\lambda + 2 = -(\lambda^3 - 3\lambda - 2)$

$\therefore -(\lambda^3 - 3\lambda - 2) = 0$

$\therefore \lambda^3 - 3\lambda - 2 = 0$

$\Rightarrow (\lambda - 2)(\lambda + 1)^2 = 0$

$\Rightarrow \lambda_1 = 2, \lambda_2 = -1, \lambda_3 = -1$

Algebraic Multiplicity of $\lambda = -1$ is 2 and of $\lambda = 2$ is 1.

Case-1: $\lambda_1 = 2$	Case-2: $\lambda_2 = -1, \lambda_3 = -1$

$$\therefore [A-\lambda I|O] = \begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array}\right] \quad R_2 \leftrightarrow R_1$$

$$= \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -2 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array}\right] \quad R_2 \to -1/3R_2$$

$$= \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 3 & -3 & 0 \end{array}\right] \quad \begin{array}{l} R_1 \to R_1 + 2R_2 \\ R_3 \to R_3 - 3R_2 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Let
$x_3 = k, x_2 - x_3 = 0, \Rightarrow x_2 = k, x_1 - x_3 = 0, x_1$
.Therefore, eigen space is for $\lambda_1 = 2$ is
$\{k(1,1,1)/k \in R\}$

$$\therefore [A-\lambda I|O] = \begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Let$x_3 = k_1, x_2 = k_2,$
$\Rightarrow x_1 + x_2 + x_3 = 0 \Rightarrow x_1 = -k_1 - k_2,.$
Therefore, eigen space is for$\lambda_2 = -1, \lambda_3 = -1$
is $\{k_1(-1,0,1) + k_2(-1,1,0)/k_1, k_2 \in R\}$
Hence, Geometric Multiplicity of$\lambda_2 = -1$ is 2 and $\lambda_1 = 2$ of is 1.

Example: Determine algebraic and geometric multiplicity of matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$.

Answer: $\lambda = 1,2,2$ therefore algebraic multiplicity of $\lambda = 2$ is 2 and geometric multiplicity is 1. For $\lambda = 1$, A.M. is 1 and G.M. is 1.

Example: Find Eigen value and Eigen vector for following matrix$\begin{bmatrix} -2 & -8 & -12 \\ 1 & 4 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

Solution: $|A-\lambda I|=0$. Also, find A.M.

$$\begin{vmatrix} -2-\lambda & -8 & -12 \\ 1 & 4-\lambda & 4 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$\therefore (-2-\lambda)((4-\lambda)\times(1-\lambda) - 4\times 0) - (-8)(1\times(1-\lambda) - 4\times 0) + (-12)(1\times 0 - (4-\lambda)\times 0) = 0$$

$$\therefore (-2-\lambda)((4-5\lambda+\lambda^2) - 0) + 8((1-\lambda) - 0) - 12(0-0) = 0$$

$$\therefore (-2-\lambda)(4-5\lambda+\lambda^2) + 8(1-\lambda) - 12(0) = 0$$

$\therefore (-8 + 6\lambda + 3\lambda^2 - \lambda^3) + (8 - 8\lambda) - 0 = 0$

$\therefore (-\lambda^3 + 3\lambda^2 - 2\lambda) = 0$
$\therefore -\lambda(\lambda - 1)(\lambda - 2) = 0$

$\therefore \lambda = 0 or (\lambda - 1) = 0 or (\lambda - 2) = 0$

$\therefore$ *The eigenvalues of the matrix A are given by* $\lambda = 0,1,2$

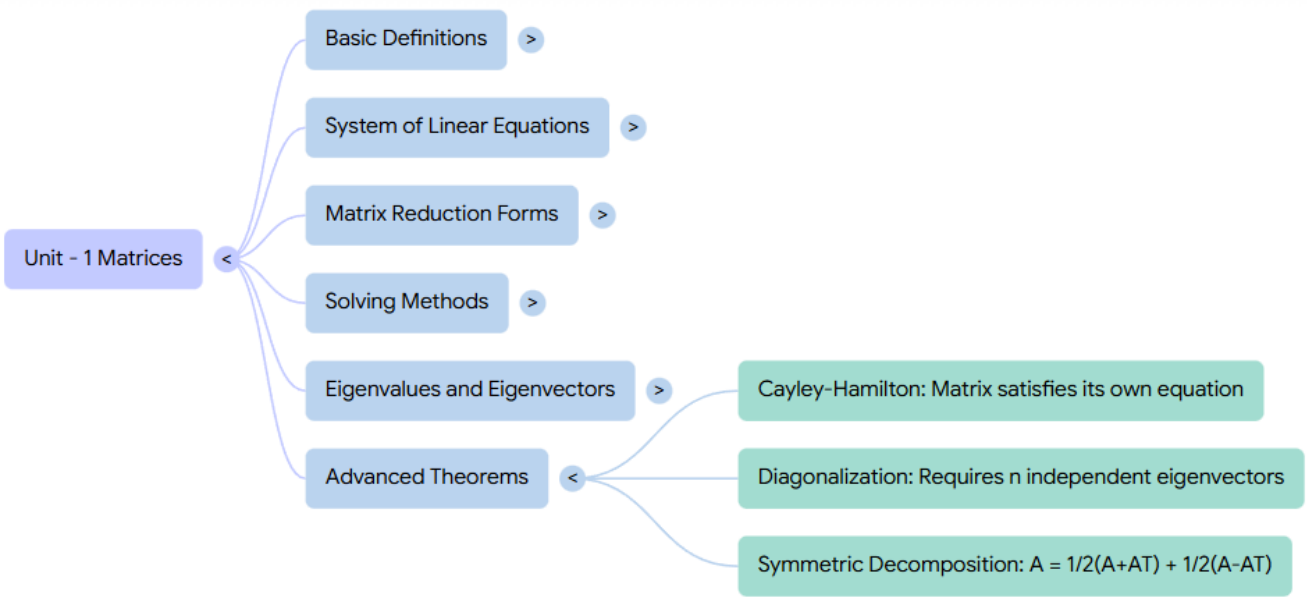
1. Eigenvectors for λ=0	2. Eigenvectors for λ=1	3. Eigenvector for λ=2
$v_1 = \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}$	$v_2 = \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$	$v_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

A.M. for all the three eigen values are 1.

Exercise-7

Find the Eigen value and Eigenvector of the matrices

a. $\begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, b. $\begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$



Schematic diagram of matrix related results

Theorem: Every square matrix can be decomposed as a sum of symmetric and skew-symmetric matrices.

Proof: Let A be $m \times n$ matrix.

Let $B = \frac{1}{2}(A + A^T)$ *and* $C = = \frac{1}{2}(A - A^T)$ be two matrices.

Obviously, $A = B + C$

Now,$B^T = \left[\frac{1}{2}(A + A^T)\right]^T = \frac{1}{2}[(A + A^T)]^T = \frac{1}{2}[A^T + (A^T)^T] = \frac{1}{2}(A^T + A) = B$

As $B^T = B$, B is symmetric.

$$C^T = \left[\frac{1}{2}(A - A^T)\right]^T = \frac{1}{2}[(A - A^T)]^T = \frac{1}{2}[A^T - (A^T)^T] = \frac{1}{2}(A^T - A) = -C$$

Therefore, $C^T = -C$, C is skew- symmetric.

Therefore, A is a sum of symmetric and skew-symmetric matrices.

1.9 Cayley –Hamilton Theorem:

Every square matrix satisfies its own characteristic equation i.e. The theorem states that, for a square matrix A of order n, if $|A - \lambda I_n| = 0$.

Example: Verify Cayley-Hamilton theorem and hence find the inverse of $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and A^4.

Solution: The characteristic equation for given matrix is

$$|A - \lambda I_2| = 0.$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(3-\lambda) - 8 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda - 5 = 0$$

Now, by putting $\lambda = A$, we have

$$A^2 - 4A - 5I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

Hence, Cayley-Hamilton theorem verified.

Now, by using Cayley-Hamilton theorem, we have

$A^2 - 4A - 5I = 0$, by applying A^{-1} on both the sides

$$A^{-1}\left(A^2 - 4A - 5I\right) = A^{-1}(0)$$

$$\Rightarrow A - 4I - 5A^{-1} = 0$$

$$\Rightarrow 5A^{-1} = A - 4I$$

$$\Rightarrow A^{-1} = \frac{1}{5}(A - 4I) = \frac{1}{5}\left(\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \frac{1}{5}\begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

And for A^4, applying A^2 both the sides

$$A^2\left(A^2 - 4A - 5I\right) = A^2(0)$$

$$\Rightarrow A^4 - 4A^3 - 5A^2 = 0$$

$$\Rightarrow A^4 = 4A^3 + 5A^2$$

$$\Rightarrow A^4 = 4\begin{bmatrix} 41 & 84 \\ 42 & 83 \end{bmatrix} + 5\begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} \Rightarrow A^4 = \begin{bmatrix} 209 & 416 \\ 208 & 417 \end{bmatrix}$$

Example: Find the characteristics equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence prove that

$$A^8-5A^7+7A^6-3A^5+A^4-5A^3+8A^2-2A+I=\begin{bmatrix}8&5&5\\0&3&0\\5&5&8\end{bmatrix}.$$

<u>Solution</u>: The characteristics equation is

$$|A-\lambda I|=\begin{vmatrix}2-\lambda&1&1\\0&1-\lambda&0\\1&1&2-\lambda\end{vmatrix}=0$$

$\therefore \lambda^3-5\lambda^2+7\lambda-3=0$

By Cayley-Hamilton Theorem

$\therefore A^3-5A^2+7A-3I=0$ .....(1)

Now,

$$A^8-5A^7+7A^6-3A^4-5A^3+8A^2-2A+I$$

$=A^5(A^3-5A^2+7A-3I)+A(A^3-5A^2+7A-3I)+A^2+A+I$

$=A^2+A+I$ $\qquad$ $u\sin g$ (1)

$$\therefore A^2+A+I=\begin{bmatrix}2&1&1\\0&1&0\\1&1&2\end{bmatrix}\begin{bmatrix}2&1&1\\0&1&0\\1&1&2\end{bmatrix}+\begin{bmatrix}2&1&1\\0&1&0\\1&1&2\end{bmatrix}+\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$=\begin{bmatrix}5&4&4\\0&1&0\\4&4&5\end{bmatrix}+\begin{bmatrix}2&1&1\\0&1&0\\1&1&2\end{bmatrix}+\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$=\begin{bmatrix}8&5&5\\0&3&0\\5&5&8\end{bmatrix}$$

Exercise:

(1)Verify Cayley-Hamilton theorem and hence find the inverse of $A=\begin{bmatrix}1&2&-2\\-1&3&0\\0&-2&1\end{bmatrix}$ and A^4.

(2) Compute $A^9-6A^8+10A^7-3A^6+A+I$, where $A=\begin{bmatrix}1&2&3\\-1&3&1\\1&0&2\end{bmatrix}$

(3) Using Cayley-Hamilton theorem find A^{-1} for the following matrix $\begin{bmatrix}2&-1&1\\-1&2&-1\\1&-1&2\end{bmatrix}$.

(4) Using Cayley-Hamilton theorem find A^{-1} for A=$\begin{bmatrix}1&-1\\2&3\end{bmatrix}$

<u>2.0 Diagonalization of a matrix:</u>

An $n\times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

OR

If $n \times n$ matrix A has a basis of eigenvectors, then $D = P^{-1}AP$ is diagonal, with the eigenvalues of A as the entries on the main diagonal. Here, P is the matrix with these eigenvectors as column vectors.
Also, $D^n = P^{-1}A^nP$ and $A^n = PD^nP^{-1}$ 3

Example-34: Find a matrix P that diagonalizes matrix A and determine $P^{-1}AP$

$$A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$$

<u>Solution :</u>

The characteristic equation is $|A - \lambda I_n| = 0$

$$\begin{vmatrix} -1-\lambda & 4 & -2 \\ -3 & 4-\lambda & 0 \\ -3 & 1 & 3-\lambda \end{vmatrix} = 0$$

$\therefore \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\therefore (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$

$\therefore \lambda = 1,2,3$

For $\lambda = 1$

$$\therefore [A - \lambda I | O] = \left[\begin{array}{ccc|c} -1-\lambda & 4 & -2 & 0 \\ -3 & 4-\lambda & 0 & 0 \\ -3 & 1 & 3-\lambda & 0 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} -2 & 4 & -2 & 0 \\ -3 & 3 & 0 & 0 \\ -3 & 1 & 2 & 0 \end{array}\right] R_1 \to -1/2R_1$$

$$= \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -3 & 3 & 0 & 0 \\ -3 & 1 & 2 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 + 3R_1 \\ R_3 \to R_3 + 3R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & -5 & 5 & 0 \end{array}\right] R_2 \to -1/3R_2$$

$$= \left[\begin{array}{ccc|c} 1 & -4/3 & 2/3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & 3 & 0 \end{array}\right] \begin{array}{l} R_1 \to R_1 + 4/3R_2 \\ R_3 \to R_3 + 3R_2 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -2/3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore z = k, y - z = 0 \;\&\; x - 2/3z = 0$

$\Rightarrow z = k, y = k, x = 2/3k$

$\therefore (x, y, z) = k(\frac{2}{3}, 1, 1); k \in R$

$\therefore (x, y, z) = 3k(2,3,3); k \in R \quad (\because 3k = k')$

$E_1 = \{k'(2,3,3) / k' \in R\}$

For $\lambda = 3$

$$\therefore [A - \lambda I | O] = \left[\begin{array}{ccc|c} -1-\lambda & 4 & -2 & 0 \\ -3 & 4-\lambda & 0 & 0 \\ -3 & 1 & 3-\lambda & 0 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} -4 & 4 & -2 & 0 \\ -3 & 1 & 0 & 0 \\ -3 & 1 & 0 & 0 \end{array}\right] R_1 \to -1/4R_1$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 1/2 & 0 \\ -3 & 1 & 0 & 0 \\ -3 & 1 & 0 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 + 3R_1 \\ R_3 \to R_3 + 3R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 1/2 & 0 \\ 0 & -2 & 3/2 & 0 \\ 0 & -2 & 3/2 & 0 \end{array}\right] R_2 \to -1/2R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -5 & 5 & 0 \end{array}\right] \begin{array}{l} R_1 \to R_1 + 2R_2 \\ R_3 \to R_3 + 5R_2 \end{array}$$

$=$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore z = k, y - z = 0 \;\&\; x - z = 0$

$\Rightarrow z = k, y = k, x = k$

$\therefore (x, y, z) = k(1,1,1); k \in R$

$E_1 = \{k(1,1,1) / k \in R\}$

For $\lambda = 2$

$$\therefore [A - \lambda I | O] = \left[\begin{array}{ccc|c} -1-\lambda & 4 & -2 & 0 \\ -3 & 4-\lambda & 0 & 0 \\ -3 & 1 & 3-\lambda & 0 \end{array}\right]$$

$=$

$$\left[\begin{array}{ccc|c} -3 & 4 & -2 & 0 \\ -3 & 2 & 0 & 0 \\ -3 & 1 & 1 & 0 \end{array}\right] R_1 \to -1/3R_1$$

$=$

$$\left[\begin{array}{ccc|c} 1 & -4/3 & 2/3 & 0 \\ -3 & 2 & 0 & 0 \\ -3 & 1 & 1 & 0 \end{array}\right] \begin{array}{l} R_2 \to R_2 + 3R_1 \\ R_3 \to R_3 + 3R_1 \end{array}$$

$=$

$$\left[\begin{array}{ccc|c} 1 & -4/3 & 2/3 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & -3 & 3 & 0 \end{array}\right] R_2 \to -1/2R_2$$

$=$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1/2 & 0 \\ 0 & 1 & -3/4 & 0 \\ 0 & -2 & 3/2 & 0 \end{array}\right] \begin{array}{l} R_1 \to R_1 + R_2 \\ R_3 \to R_3 + 2R_2 \end{array}$$

$=$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1/4 & 0 \\ 0 & 1 & -3/4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore z = k, y - 3/4z = 0 \;\&\; x - 1/4z = 0$

$\Rightarrow z = k, y = 3/4k, x = 1/4k$

$\therefore (x, y, z) = k(\frac{1}{4}, \frac{3}{4}, 1); k \in R$

$\therefore (x, y, z) = 4k(1,3,4); k \in R \quad (\because 4k = k')$

$E_1 = \{k'(1,3,4) / k' \in R\}$

$$\therefore P = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$\therefore P^{-1}AP = D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Example: Find a matrix P that diagonalizes A and determine $P^{-1}AP$where

$$A = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$$

Also find A^{10} and find eigenvalues of A^2.

$$A = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$$

<u>Solution :</u>

The characteristic equation is

$$|A - \lambda I_n| = 0$$

$$\therefore [A - \lambda I | O] = \left[\begin{array}{cc|c} 1-\lambda & 0 & 0 \\ 6 & -1-\lambda & 0 \end{array}\right]$$

$=$

$$\begin{vmatrix} 1-\lambda & 0 \\ 6 & -1-\lambda \end{vmatrix} = 0$$

$\therefore (1-\lambda)(-1-\lambda) - 0 = 0$

$\therefore \lambda = 1, -1$

For $\lambda = 1$

$$\therefore [A - \lambda I | O] = \left[\begin{array}{cc|c} 1-\lambda & 0 & 0 \\ 6 & -1-\lambda & 0 \end{array}\right]$$

$$= \left[\begin{array}{cc|c} 0 & 0 & 0 \\ 6 & -2 & 0 \end{array}\right]$$

Suppose $x = k, 6x - 2y = 0$

$x = k, y = 3k$

$\therefore (x, y) = \{k(1,3)/k \in R\}$

For $\lambda = -1$

(Continue on previous page …)

$$\left[\begin{array}{cc|c} 2 & 0 & 0 \\ 6 & 0 & 0 \end{array}\right]$$

$y = k, 6x = 0 => x = 0$

$\therefore (x, y) = \{k(0,1)/k \in R\}$

Now,

$$P = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \Rightarrow P^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = D$$

$$P^{-1}AP = D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow A = PDP^{-1}$$

$$A^{10} = PD^{10}P^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1^{10} & 0 \\ 0 & (-1)^{10} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$A^{10} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Eigenvalues of A^2 are: $1^2 = 1$ and $(-1)^2 = 1$.

1.11 Application of Matrices:

1.11.1 Coding and Encoding

Example

Consider the message to be sent: BEST WISHES

We take the standard codes as follows:

$A = 1$; $B = 2$; .....; $Z = 26$ and Space $= 0$

1. We convert the above message in to a stream of numerical values as follows:

   BEST WISHES 2 5 19 20 0 23 9 19 8 5 19

2. We construct the message matrix M with this stream of numerals as

$$M = \begin{bmatrix} 2 & 5 & 19 \\ 20 & 0 & 23 \\ 9 & 19 & 8 \\ 5 & 19 & 0 \end{bmatrix}$$ which is of order $4 X 3$

3. Then we perform the product MA, where A is an arbitrary nonsingular matrix given by

$$A=\begin{bmatrix}1 & 0 & -1\\ 0 & 2 & 2\\ 0 & 0 & 3\end{bmatrix} \text{ whose inverse is given by } A^{-1}=\begin{bmatrix}1 & 0 & \frac{1}{3}\\ 0 & \frac{1}{2} & \frac{-1}{3}\\ 0 & 0 & \frac{1}{3}\end{bmatrix}$$

4. Then we get matrix

$$X=MA=\begin{bmatrix}2 & 5 & 19\\ 20 & 0 & 23\\ 9 & 19 & 8\\ 5 & 19 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & -1\\ 0 & 2 & 2\\ 0 & 0 & 3\end{bmatrix}=\begin{bmatrix}2 & 10 & 65\\ 20 & 0 & 49\\ 9 & 38 & 53\\ 5 & 38 & 33\end{bmatrix}$$

5. $$M=XA^{-1}=\begin{bmatrix}2 & 10 & 65\\ 20 & 0 & 49\\ 9 & 38 & 53\\ 5 & 38 & 33\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{1}{3}\\ 0 & \frac{1}{2} & -\frac{1}{3}\\ 0 & 0 & \frac{1}{3}\end{bmatrix}=\begin{bmatrix}2 & 5 & 19\\ 20 & 0 & 23\\ 9 & 19 & 8\\ 5 & 19 & 0\end{bmatrix}$$

This matrix M is converted into numerical message as

2 5 19 20 0 23 9 19 8 5 19

This stream of numerical is converted into the text message as

2 5 19 20 0 23 9 19 8 5 19: *BEST WISHES*

Example

The Encoding Process

1. Convert the Message to Numbers

   The word "MATH" corresponds to the numbers: 13, 1, 20, 8.

2. Create the Message Matrix (M)

   Group the numbers into a matrix. Since there are 4 letters, we can form a 2x2 matrix.

$$M=\begin{bmatrix}13 & 20\\ 1 & 8\end{bmatrix}$$

3. Choose an Invertible Encoding Matrix (A)

   Select a square encoding matrix that has an inverse (its determinant must be non-zero).

   We will use a simple 2x2 matrix as an example.

$$A=\begin{bmatrix}2 & 1\\ 3 & 2\end{bmatrix}$$

The determinant of A is det(A) = (2 * 2) - (1 * 3) = 4 - 3 = 1.

Since the determinant is 1 (not zero), the matrix is invertible.

4. Encode the Message

Multiply the message matrix (M) by the encoding matrix (A) to get the coded matrix (C).

$$C = AXM = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 13 & 20 \\ 1 & 8 \end{bmatrix}$$

$$C = \begin{bmatrix} (2\times13)+(1\times1) & (3\times13)+(2\times1) \\ (2\times20)+(1\times8) & (3\times20)+(2\times8) \end{bmatrix} = \begin{bmatrix} 26+1 & 39+2 \\ 40+8 & 60+16 \end{bmatrix} = \begin{bmatrix} 27 & 41 \\ 48 & 76 \end{bmatrix} \quad]$$

The encoded message is the sequence of numbers from matrix C: 27, 41, 48, This is what is transmitted.

The Decoding Process

The receiver gets the sequence 27, 41, 48, 76 and knows the encoding matrix A.

1. Form the Coded Matrix (C)

The receiver arranges the numbers back into the 2x2 matrix form: $C = \begin{bmatrix} 27 & 48 \\ 41 & 76 \end{bmatrix}$

2. Find the Inverse of the Encoding Matrix

To decode, the receiver must find the inverse of the encoding matrix A.

For a 2 x 2 matrix

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is $A^{-1} = \frac{1}{\det(A)}[a * d - b * c]$

$\det(A) = 1$. The inverse is: $A^{-1} = \frac{1}{1}\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

3. Decode the Message

Multiply the coded matrix (C) by the inverse matrix (A^{-1}) to recover the original message matrix (M).

$$M = A^{-1}XC = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} X \begin{bmatrix} 27 & 48 \\ 41 & 76 \end{bmatrix}$$

$$M = \begin{bmatrix} (2X27)+(-1X41) & (2X48)+(-1X76) \\ (-3X27)+(2X41) & (-3X48)+(2X76) \end{bmatrix}$$

$$M = \begin{bmatrix} 54-41 & 96-76 \\ -81+82 & -144+152 \end{bmatrix} = \begin{bmatrix} 13 & 20 \\ 1 & 8 \end{bmatrix}$$

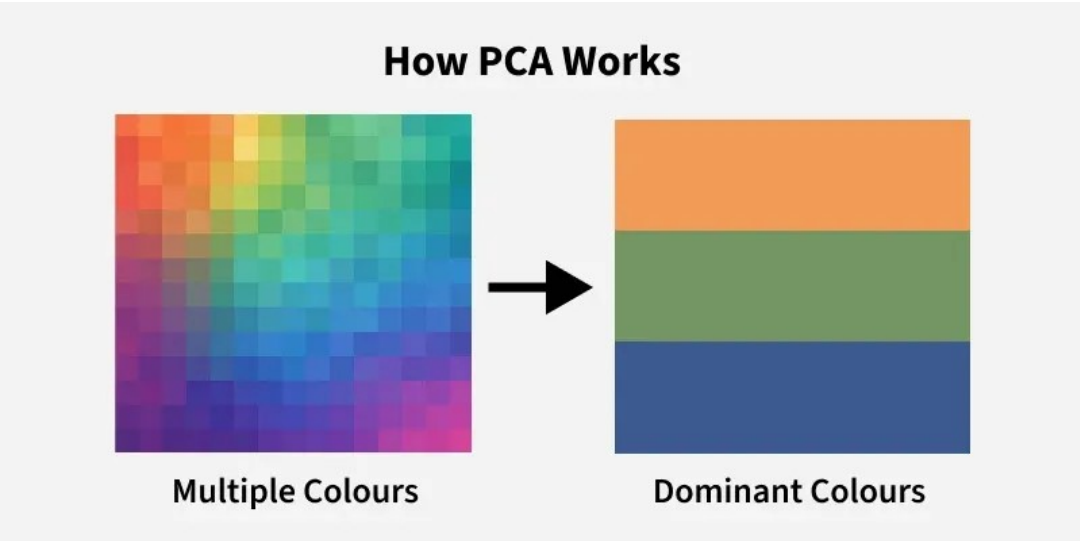
4. Convert Numbers back to the Message

The resulting numbers are 13, 1, 20, 8, which, using the correspondence table, translate back to M,

A, T, H. The original message is recovered.

1.11.2 Principal Component Analysis (PCA)

- PCA (Principal Component Analysis) is a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information.
- It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components.
- It helps us to remove redundancy, improve computational efficiency and make data easier to visualize and analyze



Principal Component Analysis (PCA)

- Principal Component Analysis (PCA) is a linear algebra–based technique used to reduce the number of variables (dimensions) in a dataset while keeping maximum information (variance).
- In simple words, PCA converts correlated variables into a new set of uncorrelated variables called principal components using eigenvalues and eigenvectors.
- Linear Algebra behind PCA is it is completely based on matrices, eigenvalues, and eigenvectors.
- Main concepts used in PCA are Mean, Covariance matrix, eigenvalues, eigen-vectors and Matrix Transformation.
- Principal components are orthogonal
- Eigenvectors = directions
- Eigenvalues = importance
- PCA is a dimension reduction technique

Need for PCA

- To reduce dimensionality of large datasets
- To remove redundancy in data
- To visualize high-dimensional data in 2D or 3D
- To speed up calculations in machine learning
- To remove noise from data

Example: Find PCA for the matrix

$$X = \begin{bmatrix} 2 & 4 \\ 4 & 6 \\ 6 & 8 \end{bmatrix}$$

Step 1: Arrange Data in Matrix Form

Suppose we have a dataset with two variables x and y:

$$X = \begin{bmatrix} 2 & 4 \\ 4 & 6 \\ 6 & 8 \end{bmatrix}$$

Each row = observation, each column = variable.

Step 2: Mean Centering the Data

Subtract mean of each column from the data:

Mean of column 1 = 4
Mean of column 2 = 6

$$X_{centered} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

Step 3: Find Covariance Matrix

Formula: $cov = \frac{1}{n-1} X_c^T X_c$

For above data: $\text{cov} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$

Step 4: Find Eigenvalues and Eigenvectors

Solve: $|Cov - \lambda I| = 0$

Eigenvalues: $\lambda_1 = 8, \quad \lambda_2 = 0$

Corresponding eigenvectors:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Interpretation of Eigenvalues and Eigenvectors

Eigenvector → direction of new axis (principal component)

Eigenvalue → amount of variance in that direction

Largest eigenvalue = **most important principal component**

Here, eigenvalue 8 is largest → keep eigenvector $\begin{bmatrix}1\\1\end{bmatrix}$

Step 5: Form Principal Components

Principal Component 1 (PC_1) is the eigenvector with highest eigenvalue: $PC_1 = \begin{bmatrix}1\\1\end{bmatrix}$

This becomes the **new axis**.

Step 6: Project Data onto Principal Component

$Z = X_{centered}\ PC_1$

This reduces **2D data → 1D data**.

Example: Given covariance matrix: $\begin{bmatrix}3 & 1\\1 & 3\end{bmatrix}$

Characteristic equation: $|A - \lambda I| = 0 \Rightarrow (3-\lambda)^2 - 1 = 0$

Eigenvalues: $\lambda_1 = 4, \quad \lambda_2 = 2$ So, **first principal component** corresponds to eigenvalue 4.

Example: Find the First Principal Component of Data Matrix: $\begin{bmatrix}1 & 2\\2 & 3\\3 & 4\end{bmatrix}$

Step 1: Mean of column 1 = 2
Mean of column 2 = 3
Mean of columns = (2, 3)

Step 2: Mean Centered Matrix:

Subtract column mean from each element: $\begin{bmatrix}1-2 & 2-3\\2-2 & 3-3\\3-2 & 4-3\end{bmatrix} = \begin{bmatrix}-1 & -1\\0 & 0\\1 & 1\end{bmatrix}$

Step 3: Covariance Matrix: $cov = \frac{1}{n-1} X_c^T X_c$

$$X_c^T = \begin{bmatrix}-1 & 0 & 1\\-1 & 0 & 1\end{bmatrix}$$

$$X_c^T X_c = \begin{bmatrix}2 & 2\\2 & 2\end{bmatrix}$$

Since n = 3, $cov = \frac{1}{3-1}\begin{bmatrix}2 & 2\\2 & 2\end{bmatrix}$

$$cov = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$$

Step 4: Find Eigenvalues

Solve: $|Cov - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$

$$(1-\lambda)^2 - 1 = 0 \Rightarrow \lambda^2 - 2\lambda = 0$$
$$\Lambda_1 = 2, \lambda_2 = 0$$

Eigenvalues: $\lambda_1 = 2, \lambda_2 = 0$
First Principal Component corresponds to λ_1.:

Example: Find PCA Using Given Covariance Matrix $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$

Given Covariance Matrix $\boldsymbol{A} = \begin{bmatrix} \mathbf{4} & \mathbf{2} \\ \mathbf{2} & \mathbf{3} \end{bmatrix}$

Solution:

$$|A - \lambda I| = 0$$
$$\begin{bmatrix} 4-\lambda & 2 \\ 2 & 3-\lambda \end{bmatrix} = 0$$
$$(4-\lambda)(3-\lambda) - 4 = 0$$
$$\lambda^2 - 7\lambda + 8 = 0$$
$$(\lambda - 4)(\lambda - 3) = 0$$
$$\lambda_1 = 4, \lambda_2 = 3$$

First principal component corresponds to $\lambda = 4$
It explains maximum variance in the data

References:

1. Elementary Linear Algebra (Textbook) By Howard Anton, Chris Rorres| Willy India Edition| 9thEdition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons.
3. Linear Algebra and Its Applications By G. Strang |Brooks/Cole India
4. Linear Algebra, By K. Hoffman and R. Kunze |Prentice Hall

Parul® University
Vadodara, Gujarat

NAAC GRADE A++

https://paruluniversity.ac.in/